

MULTIPLE WELLS IN THE SEMI-CLASSICAL LIMIT I

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0. INTRODUCTION

This paper is devoted to the Schrödinger operator, but the general problem that we are interested in could be formulated also for a formally self adjoint pseudodifferential operator

$$(0.1) \quad Pu(x, h) = (2\pi h)^{-n} \iint e^{i(x-y)\theta/h} p(x, \theta, h) u(y) dy d\theta,$$

where

$$(0.2) \quad p(x, \theta, h) \sim \sum_{j=0}^{\infty} h^j p_j(x, \theta), \quad h \rightarrow 0.$$

For large classes of such operators, Helffer and Robert [9], [10], [11], [12] have studied various types of asymptotics of eigenvalues. We are here interested in the behaviour of the eigenvalues tending to some given energy level $E_0 \in \mathbb{R}$ when $h \rightarrow 0$. Let W be the energy "surface" given by $p_0(x, \xi) = E_0$. We assume that $W = \bigcup_{j=1}^N W_j$, where W_j are disjoint and compact. These are the microlocal wells of the problem, to which the difficulties are localized, if one wants to study the eigenvalue asymptotics near E_0 , either by the direct WKB - Maslov approach (see Maslov [17], Duistermaat [5], Candelberger - Nosmas [3], Voros [26]) or by the trace formula approach combined with Fourier integral operators and functional

Most of this work was done while the first author visited the "Centre de Mathématiques, Ecole Polytechnique, Palaiseau".

calculus (see Helffer - Robert [9], [10], [11] and [12]). In particular they proved under suitable assumptions that the union of the asymptotic eigenvalues that one can construct for each well, gives the actual eigenvalues up to an error which is $\mathcal{O}(h^\omega)$.

Our problem is to study much smaller corrections of size $\exp(-(\text{const.})/h)$ due to the interaction between such wells, for instance in situations where the sequences of asymptotic eigenvalues associated to the various wells are identical. From the microlocal point of view (see [24] for instance) it is natural to expect significant result only in the case of analytic pseudodifferential operators. The situation however is quite different in the second order case of the Schrödinger operator

$$(0.3) \quad P = -h^2 \Delta + V(x)$$

where V is C^∞ , real valued and $\lim_{|x| \rightarrow \infty} V(x) > E_0$. The wells $U_1, \dots, U_N$ are then different components of $\{x; V(x) \leq E_0\}$ and U_j is the x -space projection of W_j . The reason for this is that a normalized eigenfunction $u = u_h$ with $Pu = E_h u$, $E_h \rightarrow E_0$, $h \rightarrow 0$ decays like $\exp(-(\text{const.})/h)$ on every compact set outside the wells. This has been proved in a slightly difficult context by Agmon [2] and in the semi-classical case by Simon [23]. (From the point of view of [24] this is a special case of the method of non-characteristic deformations which in general should give microlocal exponential decay results outside the W_j in the case of analytic pseudodifferential operators).

The classical example of double wells is for the quartic oscillator

$$P = -h^2 \frac{d^2}{dx^2} + x^4 - x^2$$

in one dimension and has been discussed by Landau - Lifschitz [14], §.50 exercise n° 2, Reed - Simon [20] and Harrell [7]. In addition to this case our main source of inspiration has been the work of Harrell [8] which treats the case of symmetric double wells and gives a direct way of describing the splitting of eigenvalues. In trying to generalize Harrell's approach to the case of multiple wells, we encountered a mysterious square matrix of surface integrals of the same type as in [8].

Things became much clearer when we realized that this matrix is just the off-diagonal part (up to a very small error) of the matrix of P in a suitable finite dimensional space and for a suitable basis, and finally our approach became quite different from [7], [8]. At this point we would like to mention also the interesting works of Davies [27], Mansour and Müller-Kristen [16], Jona-Lasinio, Martinelli, Scoppola [13], Simon [22], [23], Combes, Duclos, Seiler [29] and Damburg-Propin [28]. Simon [23] has obtained a general result on the splitting of the two lowest eigenvalues for double wells and we indicate at the end of this introduction the improvements given by our work.

In section 1 we develop some weighted L^2 estimates in the spirit of Agmon [2], to be applied at various places in section 2 and section 5. We also give some easy Hilbert space lemmas. In section 2 we develop the main strategy which is the following: consider the Schrödinger operator (0.3) on M , where M is $\mathbb{R}^n$ or a compact riemannian manifold. In the first case we assume that $\lim_{|x| \rightarrow \infty} (V(x) - E_0) > 0$. Let $U_1, \dots, U_N$ be the wells for the given energy E_0 . If dx^2 denotes the riemannian metric on M , then the Agmon metric $\max(V(x) - E_0, 0) dx^2$ gives a (degenerate) distance on M . We choose $M_j \subset M$ compact with smooth boundary such that $U_j \subset M_j$, $U_k \cap M_j = \emptyset$ when $k \neq j$, and let P_{M_j} be the self adjoint realisation of P in $L^2(M_j)$ with Dirichlet boundary condition. Then for normalized eigenfunctions $u = u_h$; $P_{M_j} u = E_h u$, $E_h \rightarrow E_0$, $h \rightarrow 0$, we have

$$(0.4) \quad \|u \exp(d(\cdot, U_j)/h)\|_{L^2(M_j)} + \|\nabla(u \exp(d(\cdot, U_j)/h))\|_{L^2(M_j)} = \mathcal{O}(\exp o(1)/h), \quad h \rightarrow 0.$$

Naturally we take M_j as large as possible, so that $B(U_j, S) = \{x \in M; d(x, U_j) \leq S\} \subset M_j$, where $S < S_0 = \min_{j \neq k} d(U_j, U_k)$ is close to S_0 . We now consider a sequence of intervals $I(h)$, $I(h) \rightarrow \{E_0\}$, $h \rightarrow 0$ such that P_{M_j} , P have no eigenvalues outside $I(h)$ which are "exponentially close" to $I(h)$ and write $\text{Sp}(P_{M_j}) \cap I(h) = \{\mu_{j,1}, \dots, \mu_{j,m_j}\}$ and let $\varphi_{j,k}$ be a corresponding orthonormal basis of eigenfunctions. We then prove, using only the energy decay and elementary arguments, that $I(h)$ contains exact-

ly $m_1 + \dots + m_N$ eigenvalues of P . ($I(h)$ is in the region of discrete spectrum of P . See A. Persson [21].) Moreover if $x_j \in C_0^\infty(M_j)$ is equal to 1 near $B(U_j, S)$ and $\psi_{j,k} = x_j \varphi_{j,k}$, $E = (\psi_1, \dots, \psi_{N, m_N}) \in L^2(M)$ and $F \in L^2(M)$ the space associated to $Sp(P) \cap I(h)$, then $\text{dist}(E, F) = \mathcal{O}(e^{-S/h})$ in a sense defined in section 1. We then notice that $P|_F$ can be identified with $\tau P|_E$ where τ is the projection onto E along $F^\perp$. Now τ can be approximated modulo $\mathcal{O}(e^{-S/h})$ by an "explicit" operator in terms of the $\psi_{j,k}$ and since $P(\psi_{j,k})$ are at a distance $\mathcal{O}(e^{-S/h})$ from E we get the matrix of $\tau P|_E$ in the basis of the $\psi_{j,k}$ up to an error $\mathcal{O}(e^{-2S/h})$ (Theorem 2.9). The expressions $w_{\alpha, \beta}$ in this matrix are further compared with certain surface integrals appearing in Harrell [8]. For example in a situation with many symmetries we may know that all the $\mu_{j,k}$ agree, and then asymptotic information about the $\varphi_{j,k}$ will give asymptotic information about the relative positions of the true eigenvalues of P in $I(h)$, although in this case they are all situated in an interval of length $\exp(-(S_0 + o(1))/h)$, $h \rightarrow 0$. From these arguments we also see that $Sp(P_{M_j}) \cap I(h)$ is independent of the choice of M_j modulo $\mathcal{O}(\exp(-2S/h))$ and one could also probably show that the $w_{\alpha, \beta}$ enjoy corresponding invariance properties.

The value of such general results depends very much on how explicitly the $\varphi_{j,k}$, and also to some extent the $\psi_{j,k}$, can be computed, so to get further more restrictive hypothesis on the potentials seem necessary. Many different classes are possible and in particular in the 1-dimensional case much can be done. Rather than starting with the 1-dimensional case we have preferred to look at the low eigenvalues in the case $E_0 = 0$, $V > 0$, $U_j = \{x_j\}$, $V''(x_j) > 0$. In section 3 we construct the asymptotics of the first finitely many eigenvalues and the corresponding eigenfunctions associated to each well. In the recent work [22], B. Simon has obtained these asymptotics but his results on the asymptotics of the eigenfunctions are not precise enough for our purposes. In section 4 we prove that in the real-analytic case these constructions give analytic symbols (i.e. certain growth estimates on the terms in the asymptotic expansions + remainder estimates). In section 5 we first show that the approximate eigenfunctions, which are of the form $a(x, h) \exp(-d(x, U_j)/h)$, approximate the actual eigenfunctions of P_{M_j} closely, even when the exponential

decrease factor is taken into account, in certain region $\Omega_j \subset M_j$ where $d(\cdot, U_j)$ is smooth. (In the case of a C^∞ potential, the required estimates go somewhat beyond Agmon's results [2]).

We then assume that whenever $d(U_j, U_k)$ is minimal i.e. equal to S_0 , then U_j and U_k are linked by a unique minimal geodesic $\gamma_{j,k} \subset \Omega_j \cup \Omega_k$ (or possibly a finite number of such geodesics) and that $\gamma_{j,k}$ is a non-degenerate minimum (in a very explicit way that we specify) in the space of curves joining x_j to x_k . Then, if all the $\mu_{j,k}$ have the same asymptotics (otherwise, just pass to a smaller interval $I(h)$) and $\mu_{j,k} = \mu_{j,k'}$, $\forall j, k, k'$, then the coefficients of the interaction matrix can be computed and they are of the form $b_{\alpha, \beta}(h) \exp(-S_0/h)$ where

$$b_{\alpha, \beta}(h) \sim h^{m(\alpha, \beta)} \sum_{j \in 1/2 \mathbb{N}} b_{\alpha, \beta}^{(j)} h^j.$$

Moreover, they are analytic symbols in the real-analytic case and the half degree terms disappear in the case of the principal eigenvalues.

In one (or possibly several) forthcoming paper(s) we shall study the implications of our results on the eigenvalues splitting in various situations with symmetries and for small perturbations of such situations.

In [23], Simon has proved a result on the splitting of the two lowest eigenvalues in the case of 2 non-degenerate minima for the potential under the implicit condition that each of the two eigenfunctions is sufficiently present in both wells. He shows that the splitting is then $\exp(-(S_0 + o(1))/h)$, $h \rightarrow 0$. Our proposition 2.16 gives this upper bound for the splitting in this case. Adding the geometric hypothesis mentioned above we get a much more precise result: the difference between the two lowest eigenvalues is then

$$\Delta E = \sqrt{4 \hat{w}_{1,2}^2 + (\mu_1 - \mu_2)^2} \exp(-2S/h),$$

where

$$\hat{w}_{1,2} = h^{1/2} b(h) \exp(-S_0/h)$$

and

$$b(h) \sim \sum_0^\infty b_j h^j, \quad b_0 < 0.$$

This gives a precise lower bound for the splitting, and it follows from section 6 (written partly in order to recover Simons's result completely) that this bound is still valid when the additional geometric condition is dropped.

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1. TWO PREPARATIONS

In this paper we will frequently need exponential decay estimates in the spirit of Agmon [2] (see also Carmona-Simon [4] for lower bounds). Here we first prove some a priori estimates that will make the proofs of the various decay results easy.

Theorem 1.1. *Let $\Omega \subset \mathbb{R}^n$ be a bounded open domain with C^2 boundary. Let $V \in C(\bar{\Omega}; \mathbb{R})$ and $\phi \in \text{Lip}(\bar{\Omega}; \mathbb{R})$. Then $\nabla \phi$ is well defined in $L^\infty(\Omega)$ as the almost everywhere limit of $\nabla(\chi_\varepsilon * \phi) = \chi_\varepsilon * \nabla \phi$, $\varepsilon \rightarrow 0$, where $\chi_\varepsilon \in C_0^\infty(B(0, \varepsilon))$ is a standard mollifier. For every $u \in C^2(\bar{\Omega}; \mathbb{R})$ with $u|_{\partial\Omega} = 0$ we have*

$$(1.1) \quad h^2 \int_{\Omega} \|\nabla(e^{\phi/h} u)\|^2 dx + \int_{\Omega} (V(x) - (\nabla\phi(x))^2) e^{2\phi/h} u^2 dx \\ = \int_{\Omega} e^{2\phi/h} P u(x) u(x) dx$$

where $P = -h^2 \Delta + V(x)$, $0 < h \leq 1$.

In particular if $V(x) - (\nabla\phi(x))^2 \geq 0$ a.e. and $V(x) - (\nabla\phi(x))^2 = F(x)^2$, $0 \leq F \in L^\infty(\Omega)$, then

$$(1.2) \quad h^2 \|\nabla(e^{\phi/h} u)\|^2 + \frac{1}{2} \|F e^{\phi/h} u\|^2 \leq \frac{1}{2} \left\| \frac{1}{F} e^{\phi/h} P u \right\|^2$$

where the norms are taken in $L^2(\Omega)$ (and the right hand side is defined as $+\infty$ when $\frac{1}{F} e^{\phi/h} P u \notin L^2(\Omega)$).

More generally if $V(x) - (\nabla\phi(x))^2 = F_+(x)^2 - F_-(x)^2$ a.e. where $0 \leq F_\pm \in L^\infty(\Omega)$, then

$$(1.3) \quad h^2 \|\nabla(e^{\phi/h} u)\|^2 + \frac{1}{2} \|F_+ e^{\phi/h} u\|^2 \leq \left\| \frac{1}{F_+ + F_-} e^{\phi/h} F_- u \right\|^2 + \frac{3}{2} \|F_- e^{\phi/h} u\|^2$$

Proof: Let first $\phi \in C^2(\bar{\Omega})$. Then by Green's formula

$$h^2 \int \|\nabla(e^{\phi/h} u)\|^2 dx = - \int h^2 \Delta(e^{\phi/h} u) \cdot e^{\phi/h} u dx = \\ = - \int e^{2\phi/h} h^2 \Delta(u) \cdot u dx - 2 \int e^{2\phi/h} \sum_j \frac{\partial \phi}{\partial x_j} h \frac{\partial u}{\partial x_j} u dx - \int h^2 \Delta(e^{\phi/h} u) e^{\phi/h} u dx \\ = I + II + III.$$

Here, since u^2 vanishes to the second order on the boundary,

$$II = - \int e^{2\phi/h} \sum_j \frac{\partial \phi}{\partial x_j} h \frac{\partial}{\partial x_j} (u^2) dx = \int h \frac{\partial}{\partial x_j} (e^{2\phi/h} \frac{\partial \phi}{\partial x_j}) u^2 dx = \\ = \int e^{2\phi/h} \left(2 \left(\frac{\partial \phi}{\partial x_j} \right)^2 + h \frac{\partial^2 \phi}{\partial x_j^2} \right) u^2 dx.$$

On the other hand in III we substitute

$$- h^2 \Delta(e^{\phi/h}) = - \sum e^{\phi/h} \left(\left(\frac{\partial \phi}{\partial x_j} \right)^2 + h \frac{\partial^2 \phi}{\partial x_j^2} \right),$$

so

$$h^2 \int \|\nabla(e^{\phi/h} u)\|^2 dx = - \int e^{2\phi/h} h^2 \Delta(u) u dx + \int e^{2\phi/h} (\nabla\phi)^2 u^2 dx$$

and (1.1) follows.

Now let $\phi \in \text{Lip}(\bar{\Omega})$ and extend ϕ to a Lipschitz-function on $\mathbb{R}^n$ with compact support. Put $\phi_\varepsilon = \chi_\varepsilon * \phi$. Then $\nabla \phi_\varepsilon$ is a bounded family in L^∞ and if $\nabla \phi$ is the gradient in the distribution sense, we have for all $\varphi \in C_0^\infty$:

$$\left| \int \nabla \phi \varphi dx \right| = \left| \lim_{\varepsilon \rightarrow 0} \int \nabla \phi_\varepsilon \varphi dx \right| \leq C \|\varphi\|_{L^1},$$

so $\nabla \phi \in L^\infty$. Now $\nabla \phi_\varepsilon = \chi_\varepsilon * \nabla \phi$ and according to the integration theory

$\nabla \phi_\epsilon = X_\epsilon * \nabla \phi + \nabla \phi$ a.e. when $\epsilon \rightarrow 0$. Since (1.1) holds when ϕ is replaced by ϕ_ϵ , we get (1.1) in general by the dominated convergence theorem.

To prove (1.2) it now suffices to estimate the right hand side of (1.1) by

$$\left\| \frac{1}{F} e^{\phi/h} p_u \right\| \left\| e^{\phi/h} f_u \right\| \leq \frac{1}{2} \left\| \frac{1}{F} e^{\phi/h} p_u \right\|^2 + \frac{1}{2} \left\| F e^{\phi/h} u \right\|^2.$$

To prove (1.3) we estimate the right hand side of (1.1) by

$$\begin{aligned} & \left\| \frac{1}{F} e^{\phi/h} p_u \right\|^2 + \frac{1}{4} \left\| F e^{\phi/h} u \right\|^2 \\ & \leq \frac{1}{F} \left\| e^{\phi/h} p_u \right\|^2 + \frac{1}{2} \left\| F_+ e^{\phi/h} u \right\|^2 + \frac{1}{2} \left\| F_- e^{\phi/h} u \right\|^2, \end{aligned}$$

$F = F_+ + F_-$, and notice that

$$\int (V - (\nabla \phi)^2) e^{2\phi/h} u^2 dx = \left\| F_+ e^{\phi/h} u \right\|^2 - \left\| F_- e^{\phi/h} u \right\|^2.$$

#

Remark 1.2. Let M be a compact riemannian manifold. Then dx will denote the induced volume element and ∇ the exterior differential on functions. Since we have an induced metric on $T_x^* M$ for every $x \in M$, the adjoint ∇^* is a well defined differential operator from the 1-forms to functions. Locally, if we choose an orthonormal base of C^∞ vector fields $X_1, \dots, X_n$, then $\nabla u = \sum_1^n X_j u e_j$ where $e_1, \dots, e_n$ is the dual base of 1 forms. If $\omega = \sum \omega_j e_j$ is a 1-form, it is easy to see that the formal adjoint is given by $\nabla^* \omega = \sum X_j^* (\omega_j)$. Here $X_j^* = -X_j + g_j$, where g_j is a real C^∞ function. It is obvious that if f is a function, then $\nabla^*(f\omega) = f \nabla^* \omega - \nabla f \cdot \nabla \omega$. The Laplacian on functions is now defined as $\Delta = -\nabla^* \nabla$. Theorem 1.1 still makes sense with $\mathbb{R}^n$ replaced by M if we make some obvious modifications in the discussion of $\nabla \phi$ (where it is preferable to discuss the properties of the gradient with respect to local coordinates). We claim that Theorem 1.1 still holds with these minor modifications. Indeed we only have to check the proof of (1.1) in the case when $\phi \in C^2(\bar{\Omega})$ and this proof goes through along the same lines, except that we now have to write coordinates-free formulas at each stage.

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As our second preparation, we discuss the distance between closed subspaces of a Hilbert space. Let H be a Hilbert space and let E, F be closed subspaces. Let Π_E, Π_F be the orthogonal projections on E and F respectively. We then define the non-symmetric distance $\vec{d}(E, F)$ as

$$\sup_{x \in E, \|x\|=1} d(x, F) = \left\| (I - \Pi_F) \big|_E \right\| = \left\| \Pi_E - \Pi_F \Pi_E \right\| = \left\| \Pi_E - \Pi_F \Pi_F \right\|.$$

Note that $\vec{d}(E, F) = 0$ if and only if $E \subseteq F$ and that $\vec{d}(E, G) \leq \vec{d}(E, F) + \vec{d}(F, G)$ if G is a third closed subspace of H .

Lemma 1.3. If $\vec{d}(E, F) < 1$, then $\Pi_F \big|_E : E \rightarrow F$ is injective and $\Pi_F \big|_E : F \rightarrow E$ has a bounded right inverse.

Proof : If $0 \neq x \in E$, $\Pi_F x = 0$, then $\vec{d}(E, F) = 1$ so the injectivity is clear. On the other hand if $x \in E$, we look for $y = \Pi_F z$, $z \in E$ such that $x = \Pi_E y = \Pi_E \Pi_F z$. Writing this as $x = (I - (I - \Pi_E \Pi_F))z$, we see that there is a unique solution with $\|z\| \leq \text{const.} \|x\|$. Indeed $\|(I - \Pi_E \Pi_F) \big|_E\| < 1$.

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Proposition 1.4. If $\vec{d}(E, F) < 1$ and $\vec{d}(F, E) < 1$, then $\Pi_F \big|_F : F \rightarrow E$, $\Pi_E \big|_E : E \rightarrow F$ are bijective with bounded inverses. Moreover $\vec{d}(E, F) = \vec{d}(F, E)$.

Proof : Only the last statement remains to prove. Clearly we may represent $F = \{x + Ax; x \in E\}$, where $A \in \mathcal{L}(E, E^\perp)$. Then $\Pi_E(x + Ax) = x$, and hence

$$\vec{d}(F, E)^2 = \sup_{x \in E \setminus 0} \frac{\|Ax\|^2}{\|x + Ax\|^2} = \sup_{x \in E \setminus 0} \frac{\|Ax\|^2}{\|x\|^2 + \|Ax\|^2}.$$

For $x \in E$, write $\Pi_F x = y + Ay$, $y \in E$. Then y is determined by the condition that $(y + Ay) \big|_t + \lambda t = (x \big|_t + \lambda t)$, $\forall t \in E$. Hence $y + A^* Ay = x$ so $y = (I + A^* A)^{-1} x$ and $\Pi_F x = (I + A^* A)^{-1} x + A(I + A^* A)^{-1} x$. Since $(I + A^* A)^{-1} - I = -A^* A(I + A^* A)^{-1}$ we get $\Pi_F x - x = -A^* A(I + A^* A)^{-1} x + A(I + A^* A)^{-1} x$ and hence $\|\Pi_F x - x\|^2 = \|A^* A(I + A^* A)^{-1} x\|^2 + \|A(I + A^* A)^{-1} x\|^2$. Then

$$\vec{d}(E, F)^2 = \sup_{x \in E \setminus 0} \|x\|^{-2} (\|A^* A(I + A^* A)^{-1} x\|^2 + \|A(I + A^* A)^{-1} x\|^2) =$$

$$= \sup_{y \in E \setminus 0} \|(I + A^* A)^{1/2} y\|^{-2} (\|A^* A(I + A^* A)^{-1/2} y\|^2 + \|A(I + A^* A)^{-1/2} y\|^2)$$

$$= \sup_{y \in E \setminus 0} (||y||^2 + ||Ay||^2)^{-1} ((A^*A)^2(I + A^*A)^{-1} + A^*A(I + A^*A)^{-1})y|y)$$

$$= \sup_{y \in E \setminus 0} (||y||^2 + ||Ay||^2)^{-1} ||Ay||^2 = d(F, E)^2.$$

#

2. GENERAL RESULTS ABOUT THE INTERACTION BETWEEN WELLS

In the following M will denote $\mathbb{R}^n$ or a connected n dimensional compact riemannian manifold. Let $V \in C^\infty(M; \mathbb{R})$ and put $P = -h^2\Delta + V$. This is obviously a selfadjoint operator on $L^2(M)$ when M is compact and has a discrete spectrum of finite multiplicity. Our goal is to study the behaviours of the eigenvalues of P near some fixed energy $E = E_0$, when $h \rightarrow 0$. Changing V we may suppose that $E_0 = 0$. In the case when $M = \mathbb{R}^n$ we make the following assumptions (automatically fulfilled when M is compact).

$$(2.1) \quad \text{There is an interval } I_0 =]-\infty, +a_0], \quad a_0 > 0$$

such that $V^{-1}(I_0)$ is compact.

Then P is a semi bounded symmetric operator on $C_0^\infty(M)$ and we denote by P also its Friedrichs extension. The next property is clear when M is compact and results in the case $M = \mathbb{R}^n$ from the Max-min principle, which permits to compare with known results see Roed - Simon [20], A. Persson [21] and for instance Helffer - Robert [1].

$$(2.2) \quad \text{For each } I_1 =]-\infty, a_1], \quad a_1 \leq a_0, \text{ the spectrum of } P \text{ in } I_1,$$

$Sp(P) \cap I_1$ is discrete finite and of finite multiplicity. Moreover there is a constant $N_0 \geq 0$ such that the number of eigenvalues of P in I_1 (counted with multiplicity) is less than or equal to h^{-N_0} for $0 < h \leq 1/N_0$.

Having now fixed the energy level to $E_0 = 0$ we decompose $\{x \in M; V(x) \leq 0\} = U_1 \cup \dots \cup U_N$ into finitely many compact disjoint sets $U_j \subseteq M$. The U_j are called wells. In M we consider the Agmon metric $\max\{V, 0\}dx^2$ (cf. Agmon [2]), where dx^2 denotes the Riemannian metric on M . Let $d(x, y)$ be the associated degenerate distance on M . In particular we can define the distance between wells and we put $S_0 = \min_{j \neq k} d(U_j, U_k)$. Let us fix a number

$S \in]0, S_0[$, and put $B(U_j, S) = \{x \in M; d(x, U_j) < S\}$. Then $\overline{B(U_j, S)} \cap U_k = \emptyset$ if $j \neq k$ and we can find compact manifolds with (C^2) boundary $M_j \subseteq M$ such that

$$(2.3) \quad \overline{B(U_j, S)} \subset M_j^\circ, \quad M_j \cap U_k = \emptyset \quad \text{if } k \neq j.$$

On $L^2(M_j)$ we shall consider the obvious self adjoint realisation P_{M_j} of P with Dirichlet boundary conditions. From now on h will not necessarily vary in a full interval $]0, \epsilon_0[$ but only in some subset $\mathcal{J}$ that has 0 as an accumulation point.

Let $I = [\alpha(h), \beta(h)]$, $(h \in \mathcal{J})$, be an interval with $\alpha(h), \beta(h) \rightarrow 0$, $h \rightarrow 0$ and assume that there is a function $a(h) > 0$ such that

$$(2.4) \quad |\log a(h)| = o\left(\frac{1}{h}\right), \quad h \rightarrow 0$$

$$(2.5) \quad P_{M_1}, \dots, P_{M_N} \text{ have no spectrum in the intervals}$$

$$[\alpha(h) - 2a(h), \alpha(h)], [\beta(h), \beta(h) + 2a(h)].$$

Let

$$Sp(P) \cap I = \{\lambda_1, \dots, \lambda_M\}$$

$$Sp(P_{M_j}) \cap I = \{\mu_{j,1}, \dots, \mu_{j,m_j}\}$$

and let $u_1, \dots, u_M \in L^2(M), \dots, \varphi_{j,1}, \dots, \varphi_{j,m_j} \in L^2(M_j), \dots$ be corresponding orthonormal systems. Naturally all these objects depend on h , but because of (2.2) and because of well known rough estimates on the number of eigenvalues for the Dirichlet problem, there exists a constant N_0 such that $M + m_1 + \dots + m_N = O(h^{-N_0})$.

Remark 2.1 : All our estimates are uniform with respect to additional parameters, if the various assumptions satisfy suitable uniformity conditions.

Remark 2.2 : We shall see below that the various eigenvalues $\mu_{j,1}$ are modified only by a term $O(e^{-2S/h})$ when we modify the choice of M_j . Hence the assumptions above do not really depend on the choice of the domains M_j . It is plausible that $M_j = M \setminus \bigcup_{k \neq j} U_k$ would be an optimal choice, but

the advantages of such a choice seem to be small compared with the additional technical difficulties that it would imply.

Lemma 2.3. Let $\phi_j(x) = d(x, U_j)$, $x \in M_j$. Then for every $\epsilon > 0$ we have

$$(2.6) \quad \| \nabla(e^{\phi_j/h} \varphi_{j,k}) \| + \| e^{\phi_j/h} \varphi_{j,k} \| = O(e^{\epsilon/h}), \quad h \rightarrow 0.$$

Proof : We notice that $V(x) - (\nabla \phi_j)^2 \geq 0$ in $M_j \setminus U_j$. If $\phi(x) = (1-\delta)\phi_j$, then $V(x) - (\nabla \phi)^2 \geq (2\epsilon - \delta^2)V(x)$ in $M_j \setminus U_j$. We apply (1.3) with $u = \varphi_{j,k}$, $\phi = (1-\delta)\phi_j$ and $P = -h^2 \Delta + (V - \mu_{j,k})$. Since $V(x) - \mu_{j,k} - (\nabla \phi)^2 \geq (2\delta - \delta^2)V(x) - \mu_{j,k}$, when h is sufficiently small we can choose F_+ to be a strictly positive continuous function on M_j , equal to $\sqrt{V - \mu_{j,k} - (\nabla \phi)^2}$ in the domain given by $d(x, U_j) \geq \delta$. Then F_+ in Theorem 1.1 has its support in $B(U_j, \delta)$ and (1.3) gives

$$h^2 \| \nabla(e^{\phi/h} \varphi_{j,k}) \|^2 + \| e^{\phi/h} \varphi_{j,k} \|^2 \leq C_\delta e^{2\delta/h}.$$

This also gives a bound for $\| e^{\phi/h} \nabla \varphi_{j,k} \|^2$ and we get (2.6).

#

Now let $X_j \in C_0^\infty(M_j)$ be equal to 1 in a neighborhood of $B(U_j, S)$. If we put $\psi_{j,k} = X_j \varphi_{j,k}$ then there is an $S_1 \in]S, S_0[$ such that

$$(2.7) \quad P \psi_{j,k} = \mu_{j,k} \psi_{j,k} + O(e^{-S_1/h}) \text{ in } L^2(M_j).$$

This is an easy consequence of (2.6). Let $E_j \subseteq L^2(M)$ be the subspace spanned by $\psi_{j,1}, \dots, \psi_{j,m_j}$ and let $E = \bigoplus_1^N E_j$. Let F be the space spanned by $u_1, \dots, u_N$.

Theorem 2.4. For every $\sigma < S_1$, $\vec{d}(E, F)$, $\vec{d}(F, E) = O(e^{-\sigma/h})$. Moreover, for h sufficiently small, there is a bijection $b: Sp(P) \cap I \rightarrow \bigcup_1^N (Sp(P_{M_j}) \cap I)$ such that for every $\sigma < S_1$, we have $b(\lambda) - \lambda = O(e^{-\sigma/h})$.

The main goal of this paper is to improve the second part of the theorem. We shall prove the theorem by a perturbation argument :

Proposition 2.5. Let A be a self adjoint operator in a Hilbert space H . Let $I \subseteq \mathbb{R}$ be a compact interval, $\psi_1, \dots, \psi_N \in H$ linearly independent,

$\mu_1, \dots, \mu_N \in I$ such that

$$(2.8) \quad A \psi_j = \mu_j \psi_j + r_j, \quad \| r_j \| \leq \epsilon.$$

Let $\alpha > 0$ and assume that $Sp(A) \cap (I + B(0, 2\alpha)) \setminus I = \emptyset$. Then if $E = (\psi_1, \dots, \psi_N)$ (i.e. the space spanned by $\psi_1, \dots, \psi_N$) and F is the space associated to $Sp(A) \cap I$, we have

$$(2.9) \quad \vec{d}(E, F) \leq \frac{N^{1/2} \epsilon}{\alpha (\lambda_S^{\min})^{1/2}}.$$

Here $\lambda_S^{\min}$ is the smallest eigenvalue of $S = ((\psi_j | \psi_k))$.

Remark 2.6 : If $Sp(A) \cap I$ is discrete of finite multiplicity and the right hand side of (2.9) is strictly smaller than 1, then we conclude that A has at least N eigenvalues in I .

Proof of the proposition : Let $\lambda \in \mathbb{C} \setminus (\{\mu_1, \dots, \mu_N\} \cup Sp(A))$. Then by (2.8) : $(A - \lambda) \psi_j = (\mu_j - \lambda) \psi_j + r_j$, which we write as

$$(2.10) \quad (A - \lambda)^{-1} \psi_j = (\mu_j - \lambda)^{-1} \psi_j - (\mu_j - \lambda)^{-1} (A - \lambda)^{-1} r_j.$$

If γ_R is the oriented boundary of $(I + B(0, \alpha)) \times i[-R, R]$ we have

$$\Pi_F \psi_j = \frac{1}{2\pi i} \int_{\gamma_R} (\mu_j - \lambda)^{-1} \psi_j d\lambda - \frac{1}{2\pi i} \int_{\gamma_R} (\mu_j - \lambda)^{-1} (A - \lambda)^{-1} r_j d\lambda.$$

The first integral is equal to ψ_j and the second one tends to

$$\frac{1}{2\pi i} \int_{\beta+a-i\infty}^{\beta+a+i\infty} (\mu_j - \lambda)^{-1} (A - \lambda)^{-1} r_j d\lambda - \frac{1}{2\pi i} \int_{\alpha-a-i\infty}^{\alpha-a+i\infty} (\mu_j - \lambda)^{-1} (A - \lambda)^{-1} r_j d\lambda,$$

as $R \rightarrow \infty$, where $I = [\alpha, \beta]$. With $\lambda = \beta + a + it$ or $\lambda = \alpha - a + it$ we have

$$\| (\mu_j - \lambda)^{-1} (A - \lambda)^{-1} r_j \| \leq \frac{\epsilon}{a^2 + t^2}. \text{ Hence } \| \Pi_F \psi_j - \psi_j \| \leq \frac{\epsilon}{\pi} \int_{-\infty}^{+\infty} \frac{1}{a^2 + t^2} dt = \frac{\epsilon}{a}.$$

If $u = \sum_1^N \alpha_j \psi_j \in E$, then $\| u \|^2 = (S\alpha | \alpha) \geq \lambda_S^{\min} \| \alpha \|^2$, so $\| \Pi_F u - u \| \leq$

$$\leq \sum_{j=1}^N \|\alpha_j\| \|\Pi_j \psi_j - \psi_j\| \leq \|\alpha\| N^{1/2} \frac{\varepsilon}{a} \leq \frac{N^{1/2} \varepsilon \|\alpha\|}{a(\lambda_{\min}^{1/2})}, \text{ and the proposition}$$

follows. $\#$

Proof of the theorem : In order to estimate $\vec{d}(E, F)$ we apply the proposition with $A = P$, $I = 1(h)$, $a = a(h)$, $\psi_j = \psi_{j,k}$, $u_j = u_{j,k}$. Then (2.7) gives (2.8) with $c = e^{-S_1/h}$. The number of eigenvalues μ_j is $N = \mathcal{O}(h^{-N_0})$ and since

$$(\psi_{j,k} | \psi_{j',k'}) = \delta_{(j,k), (j',k')} + \mathcal{O}(\exp - S_1/h)$$

it is clear that $\lambda_{\min}^{\min} = 1 + \mathcal{O}(e^{-\sigma/h})$ for every $\sigma < S_1$. Proposition 2.5 then gives the desired estimate on $\vec{d}(E, F)$.

In order to estimate $\vec{d}(F, E)$, let $u = u_k$ and $\lambda = \lambda_k$ for some k . We first need a decay result :

Lemma 2.7. Let $\tilde{\phi}(x) = \min_j d(x, U_j)$. Then for every $\varepsilon > 0$ we have

$$(2.11) \quad \|\nabla e^{(1-\varepsilon)\tilde{\phi}/h} u\| + \|e^{(1-\varepsilon)\tilde{\phi}/h} u\| = \mathcal{O}(e^{\varepsilon/h}), \quad h \rightarrow 0.$$

Proof : When M is compact the proof is just a minor modification of the proof of Lemma 2.3 and is left to the reader. When $M = \mathbb{R}^n$ we know from the construction of the Friedrichs extension that

$$h^2 \|\nabla u\|^2 + \int V(x) |u(x)|^2 dx = \lambda \|u\|^2.$$

At least when h is small enough, we have $|\lambda| \leq 1$ so if $\min V = -b$ we have $h^2 \|\nabla u\|^2 \leq (b+1) \|u\|^2 = b+1$. Approaching u by $\chi(\frac{x}{R})u(x)$, $R \rightarrow \infty$, where $\chi \in C_0^\infty(\mathbb{R}^n)$ is equal to 1 near 0, we see that (1.1) and hence (1.3) are valid with $\Omega = \mathbb{R}^n$, if ϕ is constant for large x . If $\chi_R(t) = t$ for $t \in [0, R]$, $\chi_R(t) = R$, for $t > R$, we put $\phi_R(x) = (1-\varepsilon)\chi_R(\tilde{\phi}(x))$. Then outside the wells we have $V(x) - \lambda - (\nabla \phi_R)^2 \geq (2\varepsilon - \varepsilon^2)V(x) - \lambda$ and this quantity is $> (2\varepsilon - \varepsilon^2)a_\delta - \lambda$ when $\tilde{\phi}(x) \geq \delta > 0$. Here $a_\delta > 0$. We apply (1.3) with $F_+ = \sqrt{V - \lambda - (\nabla \phi_R)^2}$ supported in the region $V - \lambda - (\nabla \phi_R)^2 \geq 0$ and with $F_- = \sqrt{-(V - \lambda - (\nabla \phi)^2)}$ in the complementary region :

$$h^2 \|\nabla e^{\phi_R/h} u\|^2 + \int \frac{((2\varepsilon - \varepsilon^2)V(x) - \lambda)e^{2\phi_R/h} u^2 dx}{(2\varepsilon - \varepsilon^2)V(x) - \lambda \geq 0}$$

$$\leq C \int \frac{e^{2(1-\varepsilon)\tilde{\phi}(x)/h} u^2 dx}{(2\varepsilon - \varepsilon^2)V - \lambda \leq 0}.$$

Here the right hand side is independent of R and can be estimated by $C e^{\mathcal{O}(1)/h}$ as $\varepsilon \rightarrow 0$. Letting $R \rightarrow \infty$ we get the non-trivial part of (2.11), i.e. the control of the L^2 norms on the complement of arbitrarily small neighborhoods of $\bigcup_j U_j$. $\#$

Naturally, in the compact case, we can replace $(1-\varepsilon)$ by 1 in the exponents in (2.11).

Now let $\hat{\chi}_j \in C_0^\infty(M_j)$ be equal to 1 near U_j and have disjoint supports. Define $\hat{\chi}_0 \in C_0^\infty(M)$ so that $1 = \sum_{j=0}^N \hat{\chi}_j$. With u and λ as above, we see from Lemma 2.7 that there is a constant $C_0 > 0$ such that

$$(2.12) \quad \hat{\chi}_0 u = \mathcal{O}(e^{-C_0/h}) \text{ in } L^2(M),$$

$$(2.13) \quad P \hat{\chi}_j u = \lambda \hat{\chi}_j u + \mathcal{O}(e^{-C_0/h}) \text{ in } L^2(M).$$

Applying Proposition 2.5 with $A = P_{M_j}$ we get for every $C_1 < C_0$:

$$(2.14) \quad \hat{\chi}_j u = \sum_{k=1}^{m_j} a_{jk} \psi_{jk} + \mathcal{O}(e^{-C_1/h}),$$

$$(2.15) \quad \sum_k |a_{jk}|^2 \leq \|\hat{\chi}_j u\|^2 \leq 1.$$

Possibly after decreasing C_1 we get

$$(2.16) \quad \hat{\chi}_j u = \sum_{k=1}^{m_j} a_{j,k} \psi_{jk} + \mathcal{O}(e^{-C_1/h}).$$

Using also (2.10) we get

$$(2.17) \quad u = \sum_{j=1}^N \sum_{k=1}^{m_j} a_{jk} \psi_{jk} + \mathcal{O}(e^{-C_2/h}), \quad 0 < C_2 < C_1.$$

Since the vectors u form an orthonormal basis for F we get $\vec{d}(F, E) =$

$\mathcal{O}(e^{-C_3/h})$ if $0 < C_3 < C_2$. Hence for h sufficiently small, $\vec{d}(E, F)$, $\vec{d}(F, E) < 1$ and Proposition 1.4 tells us that $\vec{d}(E, F) = \vec{d}(F, E) = \mathcal{O}(e^{-\sigma/h})$ for every $\sigma < S_1$. We also know that for h sufficiently small $N = m_1 + \dots + m_N$.

To prove the last part of the theorem let $\sigma < \sigma' < S_1$ and put $a = \exp -\sigma'/h$. We can then cover $\{\lambda_1, \dots, \lambda_N, \mu_{1,1}, \dots, \mu_{N,m_N}\}$ by intervals $I_1, \dots, I_M \subseteq I$ such that $d(I_j, I_k) \geq 2a$ if $j \neq k$ and such that $|I_j| = \mathcal{O}(h^{-N_0} a)$. Then apply Remark 2.6 with $A = P$ to each I_j . We conclude that when h is small enough, there are in each interval I_j at least as many λ 's as there are μ 's. But since $N = m_1 + \dots + m_N$ we must then have the same number of λ 's and μ 's in each I_j . Now let $b : \text{Sp}(P) \cap I \rightarrow \bigcup_j (\text{Sp}(P_{M_j}) \cap I)$ be a bijection with $\lambda \in I_j \Rightarrow b(\lambda) \in I_j$. Then $b(\lambda) - \lambda = \mathcal{O}(h^{-N_0} e^{-\sigma'/h})$ so $b(\lambda) - \lambda = \mathcal{O}(e^{-\sigma/h})$. This completes the proof of Theorem 2.4.

#

Keeping the same general hypothesis as in Theorem 2.4, it will be convenient to write $\alpha = (j, k)$, $1 \leq j \leq N$, $1 \leq k \leq m_j$, and $j(\alpha) = j$. Let Π_0 be the projection onto E along $F^\perp$ in $L^2(M)$. Π_0 is a well defined bounded operator for h sufficiently small and we have,

Lemma 2.8. For every $\sigma < S_1$ we have $\|\Pi_0 - \Pi_E\| = \mathcal{O}(e^{-\sigma/h})$.

Proof : When h is small enough we have $F = \{x + Ax; x \in E\}$, where $A : E \rightarrow E^\perp$ and $\|A\| = \mathcal{O}(e^{-\sigma/h})$, $\forall \sigma < S_1$. Then $F^\perp = \{y - A^*y; y \in E^\perp\}$ and if $z = x + y$, $x \in E$, $y \in E^\perp$, then $\Pi_E z = x$, $\Pi_0 z = \tilde{x}$, where $\tilde{x} \in E$ should verify $\tilde{x} - (x + y) \in F^\perp$ that is $\tilde{x} - (x + y) = A^*\tilde{y} - \tilde{y}$ for some $\tilde{y} \in E^\perp$. This amounts to the system $\tilde{x} - x = A^*\tilde{y}$, $y = \tilde{y}$ so $\|\tilde{x} - x\| = \mathcal{O}(e^{-\sigma/h} \|y\|)$ for every $\sigma < S_1$, which proves the lemma.

#

We next notice that $\Pi_0 = \Pi_0 \Pi_F$ and that the inverse of $\Pi_0|_F : F \rightarrow E$ is $\Pi_F|_E : E \rightarrow F$. Since F and $F^\perp$ are stable for P we have

$$(2.18) \quad \Pi_0 P \Pi_F u = \Pi_0 P u, \quad \forall u \in E.$$

Hence, if we identify E and F by means of $\Pi_0|_F$ and its inverse $\Pi_F|_E$ the operator $P|_F$ becomes $\Pi_0 P|_E$.

Theorem 2.9. For every $\sigma < S_1$ the matrix of $\Pi_0 P|_E$ for the base $\chi_{j(\alpha)} \varphi_\alpha$ is

$$\begin{pmatrix} \mu_{1,1} & & 0 \\ & \ddots & \\ 0 & & \mu_{N,m_N} \end{pmatrix} + (\omega_{\alpha,3}) + \mathcal{O}(e^{-2\sigma/h})$$

where

$$\omega_{\alpha,\beta} = h^2 \int \chi_{j(\alpha)} (\varphi_\beta \nabla \varphi_\alpha - \varphi_\alpha \nabla \varphi_\beta) \nabla \chi_{j(\beta)} dx.$$

Here we notice that $\omega_{\alpha,\beta} = \mathcal{O}(e^{-\sigma/h})$ for every $\sigma < S_1$ and that $\omega_{\alpha,\beta} = \mathcal{O}(e^{-2\sigma/h})$ when $j(\alpha) \neq j(\beta)$.

Proof : Recalling the notation $\psi_{j,k} = \chi_{j(\alpha)} \varphi_\alpha$, $\alpha = (j,k)$, we have

$$\Pi_E u = \sum_{\alpha,\beta} c_{\alpha,\beta} (u | \psi_\beta) \psi_\alpha,$$

where the constants $c_{\alpha,\beta}$ are determined by the condition

$$\sum_{\alpha,\beta} c_{\alpha,\beta} (\psi_\gamma | \psi_\beta) \psi_\alpha = \psi_\gamma, \quad \forall \gamma,$$

that is

$$\sum_{\alpha,\beta} c_{\alpha,\beta} (\psi_\gamma | \psi_\beta) = \delta_{\alpha,\gamma}.$$

Hence $C = (c_{\alpha,\beta}) = {}^t S^{-1}$, where $S = ((\psi_\alpha | \psi_\beta))$. Now $S = I + \mathcal{O}(e^{-\sigma/h})$ for every $\sigma < S_1$ so if we define $\tau u = \sum_{\alpha} (u | \psi_\alpha) \psi_\alpha$, then $\|\tau - \Pi_E\| = \mathcal{O}(e^{-\sigma/h})$ and the preceding lemma shows that

$$(2.19) \quad \|\tau - \Pi_0\| = \mathcal{O}(e^{-\sigma/h}) \quad \text{for every } \sigma < S_1.$$

We next note that

$$P(\psi_\beta) = \mu_\beta \psi_\beta - h^2 (2 \nabla \chi_{j(\beta)} \nabla \varphi_\beta + (\Delta \chi_{j(\beta)}) \varphi_\beta).$$

Here the last term is $\mathcal{O}(e^{-\sigma/h})$ for every $\sigma < S_1$ so by (2.19) :

$$\begin{aligned}
(2.20) \quad \Pi_0 P(\psi_\beta) &= \mu_\beta \psi_\beta - h^2 \tau (2 \nabla x_{j(\beta)} \nabla \varphi_\beta + (\Delta x_{j(\beta)}) \varphi_\beta) + \mathcal{O}(e^{-2\sigma/h}) \\
&= \mu_\beta \psi_\beta - h^2 \int_\alpha \int 2 \nabla x_{j(\beta)} \nabla \varphi_\beta x_{j(\alpha)} \varphi_\alpha + (\Delta x_{j(\beta)}) \varphi_\beta x_{j(\alpha)} \varphi_\alpha dx \psi_\alpha \\
&\quad + \mathcal{O}(e^{-2\sigma/h}).
\end{aligned}$$

If we integrate by parts the second term in the integral, then the term containing $\nabla x_{j(\beta)} \nabla x_{j(\alpha)}$ may be omitted and we get

$$\begin{aligned}
(2.21) \quad \Pi_0 P(\psi_\beta) &= \\
&= \mu_\beta \psi_\beta + h^2 \int_\alpha \int x_{j(\alpha)} (\varphi_\beta \nabla \varphi_\alpha - \varphi_\alpha \nabla \varphi_\beta) \cdot \nabla x_{j(\beta)} dx \psi_\alpha + \mathcal{O}(e^{-2\sigma/h})
\end{aligned}$$

for every $\sigma < S_1$.

¶

We notice that $\Pi_0 P|_E = \Pi_E P|_E + \mathcal{O}(e^{-2\sigma/h})$, $\forall \sigma < S_1$, and $\Pi_E P|_E$ is self adjoint. However we cannot expect the matrix $(w_{\alpha,\beta})$ to be symmetric because the ψ_α do not form an orthonormal basis in E . We have

$$\begin{aligned}
w_{\alpha,\beta} &= h^2 \int \nabla x_{j(\beta)} \nabla \varphi_\alpha x_{j(\alpha)} \varphi_\beta - \nabla x_{j(\beta)} \nabla \varphi_\beta x_{j(\alpha)} \varphi_\alpha dx = \\
&= h^2 \int -x_{j(\beta)} \Delta \varphi_\alpha x_{j(\alpha)} \varphi_\beta - x_{j(\beta)} \nabla \varphi_\alpha \nabla x_{j(\alpha)} \varphi_\beta - x_{j(\beta)} x_{j(\alpha)} \nabla \varphi_\alpha \nabla \varphi_\beta \\
(2.22) \quad &+ x_{j(\beta)} \Delta \varphi_\beta x_{j(\alpha)} \varphi_\alpha + x_{j(\beta)} \nabla \varphi_\beta \nabla x_{j(\alpha)} \varphi_\alpha + x_{j(\beta)} x_{j(\alpha)} \nabla \varphi_\beta \nabla \varphi_\alpha dx \\
&= (\mu_\alpha - \mu_\beta) \int \psi_\alpha \psi_\beta dx + w_{\beta,\alpha} = (\mu_\alpha - \mu_\beta) t_{\alpha,\beta} + w_{\beta,\alpha}.
\end{aligned}$$

Here the matrix $T = (t_{\alpha,\beta})$ is defined by $S = I + T$, so $T = \mathcal{O}(e^{-\sigma/h})$, $\forall \sigma < S_0$. If $\vec{\psi} = (\psi_1, \dots, \psi_N, m_N)$ then $\vec{e} = \vec{\psi} S^{-1/2}$ is an O.N. basis in E (that we call the orthonormalization of $\vec{\psi}$). The matrix of $\Pi_0 P|_E$ in this basis is

$$\begin{aligned}
&\equiv (I+T)^{1/2} ((\mu_\alpha \delta_{\alpha,\beta}) + (w_{\alpha,\beta})) (I+T)^{-1/2} \equiv (I + \frac{1}{2}T) ((\mu_\alpha \delta_{\alpha,\beta}) + (w_{\alpha,\beta})) (I - \frac{1}{2}T) \equiv \\
&\equiv (\mu_\alpha \delta_{\alpha,\beta} + w_{\alpha,\beta}) + \frac{1}{2} [T, (\mu_\alpha \delta_{\alpha,\beta})] \equiv (\mu_\alpha \delta_{\alpha,\beta}) + \tilde{w}_{\alpha,\beta},
\end{aligned}$$

where $\tilde{w}_{\alpha,\beta} = w_{\alpha,\beta} + \frac{1}{2} t_{\alpha,\beta} (\mu_\beta - \mu_\alpha)$ and $\equiv$ means equivalence modulo a term

which is $\mathcal{O}(e^{-2\sigma/h})$ for every $\sigma < S_1$. Since $t_{\alpha,\beta}$ is symmetric the same is true for $\tilde{w}_{\alpha,\beta}$ by (2.22) and we have $\tilde{w}_{\alpha,\beta} = \frac{1}{2} (w_{\alpha,\beta} + w_{\beta,\alpha})$. We get

Theorem 2.10. Let $S = ((\psi_\alpha | \psi_\beta))$ and let $\vec{e} = \vec{\psi} S^{-1/2}$ be the orthonormalization of the basis $\vec{\psi}$. Then for every $\sigma < S_1$ the matrix of $\Pi_0 P|_E$ with respect to $\vec{e}$ is

$$(2.23) \quad \begin{pmatrix} \mu_1, 1 & & 0 \\ & \ddots & \\ 0 & & \mu_N, m_N \end{pmatrix} + (\hat{w}_{\alpha,\beta}) + \mathcal{O}(e^{-2\sigma/h}).$$

Here,

$$\begin{aligned}
\hat{w}_{\alpha,\beta} &= \frac{1}{2} (w_{\alpha,\beta} + w_{\beta,\alpha}) = \\
&= \frac{h^2}{2} \int x_{j(\alpha)} (\varphi_\beta \nabla \varphi_\alpha - \varphi_\alpha \nabla \varphi_\beta) \nabla x_{j(\beta)} + x_{j(\beta)} (\varphi_\alpha \nabla \varphi_\beta - \varphi_\beta \nabla \varphi_\alpha) \nabla x_{j(\alpha)} dx.
\end{aligned}$$

Remark 2.11 : Let $A = \{(j,k) ; 1 \leq j \leq N, 1 \leq k \leq m_k\}$ and write $\vec{e} = \{e_\alpha\}_{\alpha \in A}$.

The preceding theorem also gives the matrix of $P|_E$ in the basis $f_\alpha = \Pi_E(e_\alpha)$, $\alpha \in A$. Since $e_\alpha = f_\alpha + r_\alpha$ with $r_\alpha \in F^\perp$, $\|r_\alpha\| = \mathcal{O}(e^{-\sigma/h})$, $\forall \sigma < S_1$, we have

$$(f_\alpha | f_\beta) = (e_\alpha | e_\beta) + (r_\alpha | r_\beta) = \delta_{\alpha,\beta} + \mathcal{O}(e^{-2\sigma/h}).$$

If we replace $\{f_\alpha\}_{\alpha \in A}$ by its orthonormalization, it is then clear that the matrix of $P|_E$ is still of the form (2.23) in this new basis.

It is easy to compare the eigenvalues of two symmetric matrices of size N_0 whose difference is $\mathcal{O}(e^{-2\sigma/h})$ for every $\sigma < S_1$ and we get

Theorem 2.12. For h sufficiently small, there is a bijection $b : Sp(P|_E) \rightarrow Sp((\mu_\alpha \delta_{\alpha,\beta} + \hat{w}_{\alpha,\beta}))$ such that $|b(\lambda) - \lambda| = \mathcal{O}(e^{-2\sigma/h})$ for every $\sigma < S_1$.

It should be pointed out that in the case of the lowest eigenvalues, if we apply the formal L.C.A.O. method well known by chemists, using the ψ_α as "approximate" eigenfunctions, then we get precisely the matrix (2.23). Our results therefore extend and make rigorous the results of the L.C.A.O. method.

Remark 2.13 : Naturally our results are more interesting when the $\mu_\alpha - \mu_\beta$ are small. Assume that $0 < a \leq S_0$, $S_0 + a < 2S_1$. We suppose that for every $\varepsilon > 0$,

$$(2.24) \quad \mu_\alpha - \mu_\beta = \mathcal{O}(e^{-(a-\varepsilon)/h})$$

Then by (2.22)

$$w_{\alpha,\beta} \equiv w_{\beta,\alpha} \text{ mod } (e^{-(S_0+a-\varepsilon)/h}), \quad \forall \varepsilon > 0$$

and modulo errors of the same size, $w_{\alpha,\beta}$ does not depend on the choice of $\chi_{j(\alpha)}$, $\chi_{j(\beta)}$ (while $S_1 > S$ does depend on these cut off functions). If $d(U_{j(\alpha)}, U_{j(\beta)}) \geq S_0 + a$ or $j(\alpha) \neq j(\beta)$ then $w_{\alpha,\beta} \equiv 0$, otherwise let E be the closed "ellipse" $d(U_{j(\alpha)}, x) + d(U_{j(\beta)}, x) \leq S_0 + a$, so that $E \subset \dot{M}_{j(\alpha)} \cup \dot{M}_{j(\beta)}$.

Let $\Omega \subset M$ be open with smooth boundary such that $U_{j(\alpha)} \subset \Omega$, $U_{j(\beta)} \cap \bar{\Omega} = \emptyset$ and such that $E \cap \bar{\Omega} \subset \dot{M}_{j(\alpha)}$, $E \cap \Omega^c \subset \dot{M}_{j(\beta)}$. Then $\Gamma = \partial\Omega \cap E$ is a compact subset of $\dot{M}_{j(\alpha)} \cap \dot{M}_{j(\beta)}$. We claim that for every $\varepsilon > 0$

$$(2.25) \quad w_{\alpha,\beta} \equiv h^2 \int_{\Gamma} \varphi_\alpha \frac{\partial \varphi_\beta}{\partial n} - \varphi_\beta \frac{\partial \varphi_\alpha}{\partial n} dS \text{ mod } \mathcal{O}(e^{-(S_0+a-\varepsilon)/h})$$

where n is the interior normal with respect to Ω .

In fact, let $\chi_E \in C_0^\infty(M)$ be equal to 1 on E and with support close to E . Then $\text{supp } \chi_E \cap \bar{\Omega} \subset \dot{M}_{j(\alpha)}$, $\text{supp } \chi_E \cap \Omega^c \subset \dot{M}_{j(\beta)}$ and we may choose $\chi_{j(\alpha)}$, $\chi_{j(\beta)}$ equal to 1 on $\text{supp } \chi_E \cap \bar{\Omega}$ and $\text{supp } \chi_E \cap \Omega^c$ respectively. Then

$$\begin{aligned} w_{\alpha,\beta} &\equiv h^2 \int \chi_E \chi_{j(\alpha)} (\varphi_\alpha \nabla \varphi_\beta - \varphi_\beta \nabla \varphi_\alpha) \nabla \chi_{j(\beta)} dx = \\ &= h^2 \int_{\Omega} \chi_E (\varphi_\alpha \nabla \varphi_\beta - \varphi_\beta \nabla \varphi_\alpha) \nabla \chi_{j(\beta)} dx. \end{aligned}$$

Here we use the following version of Stokes' formula. If $\bar{\Omega}$ is a compact Riemannian manifold with boundary, ω a 1 form on $\bar{\Omega}$ and u a function on $\bar{\Omega}$ (all smooth up to the boundary) then

$$\int_{\Omega} \omega \cdot \nabla u dx = \int_{\Omega} (\nabla^* \omega) u dx - \int_{\partial M} \langle \omega, \frac{\partial}{\partial n} \rangle u dS.$$

Applying this, we may neglect the terms with $\nabla \chi_E$ and since $\nabla^*(\varphi_\alpha \nabla \varphi_\beta - \varphi_\beta \nabla \varphi_\alpha) = -\varphi_\alpha \Delta \varphi_\beta + \varphi_\beta \Delta \varphi_\alpha \equiv 0$, we get (2.25).

The same arguments can be applied with rapidly decreasing errors. Assume that for every $N \in \mathbb{N}$:

$$(2.24') \quad \mu_\alpha - \mu_\beta = \mathcal{O}(h^N)$$

and that we have an improved energy decay result :

(2.26) For every α and every compact set $K \subset \dot{M}_{j(\alpha)}$, there is a constant N_0 such that

$$\|\nabla(e^{\phi_{j(\alpha)}/h} \varphi_\alpha)\|_{L^2(K)} + \|e^{\phi_{j(\alpha)}/h} \varphi_\alpha\|_{L^2(K)} = \mathcal{O}(h^{-N_0}).$$

Then, if $d(U_{j(\alpha)}, U_{j(\beta)}) > S_0$ or if $\alpha \neq \beta$, we have $w_{\alpha,\beta} = \mathcal{O}(h^N e^{-S_0/h})$ for every $N \in \mathbb{N}$. Otherwise we choose $a > 0$ very small and define Γ as above. Then

$$(2.25') \quad w_{\alpha,\beta} \equiv h^2 \int_{\Gamma} \varphi_\alpha \frac{\partial \varphi_\beta}{\partial n} - \varphi_\beta \frac{\partial \varphi_\alpha}{\partial n} dS + \mathcal{O}(h^N e^{-S_0/h})$$

$\forall N \in \mathbb{N}$. In this situation we also notice that

$$w_{\alpha,\beta} \equiv w_{\beta,\alpha} \equiv \hat{w}_{\alpha,\beta} \text{ mod } \mathcal{O}(h^N e^{-S_0/h})$$

for $\forall N \in \mathbb{N}$.

The formulas (2.25), (2.25') should be compared with that of Harrell [8] Theorem 2.1.

Corollary 2.14. *In the special case when there is only one well U_1 , then for h sufficiently small there is a bijection $b: Sp(P) \cap I(h) \rightarrow Sp(P_{M_1}) \cap I(h)$ such that $|b(\lambda) - \lambda| = \mathcal{O}(e^{-2\sigma/h})$ for every $\sigma < S_1$.*

In fact, in this case we have $\hat{w}_{\alpha,\beta} = \mathcal{O}(e^{-2\sigma/h})$ as already noticed. We also notice that all our discussion so far is valid for the Dirichlet problem when M is a compact riemannian manifold with boundary and when $M_j \subset \dot{M}$.

Returning now to the general situation let us examine the effect on $\text{Sp}(P_{M_1})$ if we replace M_1 by $\tilde{M}_1 \subseteq M$ having the same properties, that is $\tilde{M}_1$ is a compact manifold with boundary, containing $B(U_1, S_1)$ in its interior and disjoint from $U_2, \dots, U_N$. Here $S_1 < S_{(1)} = \min_{\text{def } k \neq 1} d(U_1, U_k)$.

Proposition 2.15. *Let $c(h) \in]0, a(h)[$, $\log c(h) = o(\frac{1}{h})$. Then for h sufficiently small, there is a bijection $b: \text{Sp}(P_{M_1}) \cap I(h) \rightarrow \text{Sp}(\tilde{P}_{M_1}) \cap (I(h) + \overline{B(0, c(h))})$ such that $|b(\lambda) - \lambda| = \mathcal{O}(e^{-2\sigma/h})$ for every $\sigma < S_1$.*

Proof: Introducing a third domain M_1 , we can reduce the proof to the case when $M_1 \subseteq \tilde{M}_1$. Let $\mathcal{J}(h)$ be a family of intervals, having the same properties as $I(h)$ with respect to $\text{Sp}(P)$, $\text{Sp}(P_{M_j})$ and $\text{Sp}(P_{M_1})$, and such that $I(h) + B(0, 2a(h)) \subseteq \mathcal{J}(h)$. Corollary 2.14 and the subsequent observation give a bijection $\text{Sp}(P_{M_1}) \cap \mathcal{J}(h) \rightarrow \text{Sp}(\tilde{P}_{M_1}) \cap \mathcal{J}(h)$ and it suffices to restrict this bijection to $I(h) \cap \text{Sp}(P_{M_1})$.

#

We now define the "spectrum" $\text{Sp}(U_1)$ as the collection of all the $\text{Sp}(\tilde{P}_{M_1}) \cap (I(h) + \overline{B(0, c(h))})$ as in the proposition. If $\mathcal{J} \ni h \rightarrow \lambda(h) \in \mathbb{R}$ and $\sigma < S_{(1)}$ then we say that $d(\lambda, \text{Sp}(U_1)) = \mathcal{O}(e^{-2\sigma/h})$ if the distance from $\lambda(h)$ to $\text{Sp}(\tilde{P}_{M_1}) \cap (I(h) + \overline{B(0, c(h))})$ is $\mathcal{O}(e^{-2\sigma/h})$, whenever we chose $\tilde{M}_1$ with $B(U_1, c)$ in its interior. Naturally we shall use the same terminology for the other wells $U_2, \dots, U_N$.

Let $\mathcal{J} \ni h \rightarrow u(h) \in \mathcal{D}_P$, $\|u\| = 1$, $(P - \lambda)u = 0$ where $\lambda = \lambda(h) \in I(h)$. We consider functions ψ on M such that

$$(2.27) \quad |\psi(x) - \psi(y)| \leq d(x, y), \quad \forall x, y \in M,$$

with ψ constant on each U_j , and such that

$$(2.28) \quad \text{For every } \epsilon > 0 \text{ and compact set } K \subset M;$$

$$\|\nabla(e^{\psi/h} u)\|_{L^2(K)} + \|e^{\psi/h} u\|_{L^2(K)} = \mathcal{O}(e^{\epsilon/h}), \quad h \rightarrow 0.$$

If ψ_1, ψ_2 are such functions $\max(\psi_1, \psi_2)$ is another one. It is then clear that there is a largest function ψ satisfying (2.27), (2.28). We assume that the U_j have diameter 0 for the Agmon metric and that

(2.29) ψ is largest function satisfying (2.27), (2.28) and no larger function can be found even if we replace $\mathcal{J}$ by any other subset having 0 as accumulation point.

If we introduce $a_j = \psi|_{U_j}$ (which is a constant ≥ 0), then at least one of the a_j is equal to 0 and $|a_j - a_k| \leq d(U_j, U_k)$. Moreover, as in the proof of Lemma 2.5 and Lemma 2.7 we see that ψ is

$$(2.30) \quad \psi(x) = \min_j (d(x, U_j) + a_j).$$

For each j , let r_j be the largest number in $[0, S_j]$ such that $\psi(x) = a_j + d(x, U_j)$ for $x \in B(U_j, r_j)$. If $a_j = 0$ then $r_j \geq \frac{1}{2} S_0$. Consider a j for which $r_j > 0$. Then $a_j + r_j \leq a_k + d(x, U_k)$ for all $k \neq j$ and $x \in \partial B(U_j, r_j)$. Hence $a_j + 2r_j \leq a_k + d(U_j, x) + d(x, U_k)$ for all $x \in \partial B(U_j, r_j)$ and varying x we find $a_j + 2r_j \leq a_k + d(U_j, U_k)$ for all $k \neq j$. Now, for every $\epsilon > 0$ there exists a constant C_ϵ such that

$$(2.31) \quad \frac{1}{C_\epsilon} e^{-\epsilon/h} \leq e^{a_j/h} (\| \nabla u \|_{L^2(B(U_j, r_j))} + \| u \|_{L^2(B(U_j, r_j))}) \leq C_\epsilon e^{\epsilon/h}.$$

In fact, the inequality to the right follows from (2.28) and if the inequality to the left were false, then after passing to a subsequence in h we would be able to improve the choice of ψ , contrary to the assumption (2.29). Standard a priori estimates for the Laplace operator show that we can in fact eliminate the gradient term in (2.31).

Proposition 2.16. *Under the above assumptions we have for every $\epsilon > 0$:*
 $d(\lambda, \text{Sp}(U_j)) = \mathcal{O}(e^{-2(r_j - \epsilon)/h})$.

Proof: Let $\epsilon > 0$ and choose $x \in C_0^\infty(B(U_j, r_j))$ equal to 1 in a neighborhood of $\overline{B(U_j, r_j - \epsilon)}$. Let $\tilde{u} = \|u\|^{-1} u$ where all L^2 norms now are taken over $B(U_j, r_j)$. Let $v = x \tilde{u}$. Then $\|\tilde{u}\| = 1$ and $\|\nabla(e^{d(x, U_j)} \tilde{u})\| + \|e^{d(x, U_j)} \tilde{u}\| = \mathcal{O}(e^{\alpha/h})$ for every $\alpha > 0$. We have

$$(2.32) \quad P v = \lambda v + \mathcal{O}(e^{-(r_j - \epsilon)/h}) \quad \text{in } L^2(B(U_j, r_j)).$$

Let M_j be as usual and with $\text{supp } x$ in its interior. Let $F \subseteq L^2(M_j)$ be the

sum of the eigenspaces associated to $\text{Sp}(P_{M_j}) \cap I(h)$. As in the proof of Proposition 2.5 we see that $d(v, F) = \mathcal{O}(e^{-(r_j - \epsilon)/h})$, hence $\|\pi_F v\| \equiv \|v\| \equiv 1$ modulo exponentially small errors. Now $P \pi_F v = \lambda \pi_F v + \pi_F [P, X]u$, and if we explicit π_F by means of an O.N. basis of eigenvectors (on which we apply our decay estimates) we see that $\|\pi_F [P, X]u\| = \mathcal{O}(e^{-2(r_j - \epsilon)/h})$. It is then clear that $d(\lambda, \text{Sp}(U_j)) = \mathcal{O}(e^{-2(r_j - \epsilon)/h})$. $\#$

If $a_j = 0$ then $r_j \geq \frac{1}{2} \min_{k \neq j} (d(U_j, U_k)) = \frac{1}{2} S_j$ and the Proposition shows that

$$d(\lambda, \text{Sp}(U_j)) = \mathcal{O}(e^{-(S_j - \epsilon)/h})$$

for every $\epsilon > 0$. On the other hand, if for some j

$$d(\lambda, \text{Sp}(U_j)) \geq \frac{1}{C} e^{-\epsilon/h},$$

for every $\epsilon > 0$, then $r_j = 0$. We then say that U_j is a non resonant well. In a future paper we plan to study the interaction of a group of wells "through" a second group of non resonant wells.

3. FORMAL CONSTRUCTIONS NEAR A STRICT MINIMUM OF THE POTENTIAL

The computations of this section are in principle well known. They correspond to the symplectic case in the study of operators with double characteristics. B. Simon [22] has very recently given results in the present frame work, showing that the lowest eigenvalues in the case of one point-well, where V has a strict minimum, admit asymptotic expansions in powers of h . It is not true however, that only integer powers of h appear in these expansions. We shall also need more precise information about the eigenfunctions and in this section and the following we follow our original plan to use a canonical transformation. This transformation is not absolutely necessary but we feel that the computations become simpler and we can for instance examine quite explicitly when half exponents appear in the asymptotic expansions of the eigenvalues. In the next section we shall prove that in the analytic case the asymptotic expansions are actually analytic symbols and in section 5 we combine our results with decay estimates to establish the link between the formal eigenvalues and the actual ones, both in the analytic and C^∞ case.

The general assumption in this and the following two sections is that we have fixed one M_j as before and that the corresponding well is reduced to a point $U_j = \{x_0\}$. The potential V then has a minimum at x_0 and we assume that this minimum is strict

$$(3.1) \quad V''(x_0) > 0.$$

To our operator P we associate the symbol $p(x, \xi) = \xi^2 + V(x)$, where ξ^2 stands for $\|\xi\|^2$ (extended holomorphically) in the riemannian case. From the point of view of Fourier integral operators, our weights $e^{\psi(x)/h}$ (or various real valued functions $\psi(x)$) correspond to the complex phase functions $i\psi(x)$. It is therefore natural (profiting in a very essential way from the fact that we study a second order operator) to consider the symbol $-p(x, i\xi) = \xi^2 - V(x) = q(x, \xi)$. Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of the quadratic form $\frac{1}{2} \langle V''(x_0)t, t \rangle$ with respect to the riemannian metric. Then we can choose coordinates centered at x_0 so that

$$(3.2) \quad q(x, \xi) = \xi^2 - \sum_{j=1}^n \lambda_j x_j^2 + \mathcal{O}(|x, \xi|^3),$$

where now $\xi^2 = \sum_{j=1}^n \xi_j^2$ has the usual sense. The Hamilton field is then

$$(3.3) \quad H_q = 2 \left(\xi \frac{\partial}{\partial x} + \sum_{j=1}^n \lambda_j x_j \frac{\partial}{\partial \xi_j} \right) + \mathcal{O}(|x, \xi|^2).$$

The flow vanishes at $(0, 0)$ and its linearization is given by the fundamental matrix

$$(3.4) \quad F = 2 \begin{pmatrix} 0 & & & I \\ \lambda_1 & 0 & & \\ 0 & & \lambda_n & \\ & & & 0 \end{pmatrix}$$

The eigenvalues of F are $\pm 2\sqrt{\lambda_j}$ and the eigenspaces corresponding to positive respectively negative eigenvalues are

$$(3.5) \quad \Lambda_\pm^0 : \xi_j = \pm \sqrt{\lambda_j} x_j.$$

The $\Lambda_\pm^0$ are Lagrangian subspaces of $T_{(0,0)}(T^*M_1)$ of the form

$$(3.6) \quad \xi = \pm \phi'(x),$$

where in the chosen coordinates $\phi(x) = \sum_{j=1}^n \frac{1}{2} \sqrt{\lambda_j} x_j^2$. According to well known results on stable manifolds (see Abraham-Marsden [1]) there exist in a suitable neighborhood W of $(0,0)$ unique closed connected C^∞ manifolds $\Lambda_\pm$ of dimension n such that

$$(3.7) \quad T_{(0,0)}(\Lambda_\pm) = \Lambda_\pm^0, \quad H_q \text{ is tangent to } \Lambda_\pm.$$

Moreover

$$(3.8) \quad \Lambda_\pm = \{(x, \xi) \in W; \exp(-t H_q)(x, \xi) \rightarrow (0,0) \text{ for all } t \in [0, +\infty[\}$$

and if $(x, \xi) \in \Lambda_\pm$ then $\exp(-t H_q)(x, \xi) \rightarrow (0,0)$ exponentially fast when $t \rightarrow \pm\infty$. These results also hold in the real analytic category, and we gave a simple proof in the appendix of [15] of this probably well known fact. A simple argument given in the same paper shows that $\Lambda_\pm$ are Lagrangian, hence of the form

$$(3.9) \quad \Lambda_\pm : \xi = \pm \psi'(x), \quad x \in \Omega$$

where Ω is a suitable neighborhood of 0. (Here we decrease W if necessary and we can then take $W = T^*M|_\Omega$). Here ψ is a C^∞ (resp. analytic) function uniquely determined if we impose that $\psi(0) = 0$. We also have $\psi'(0) = 0$.

Let $[a, b] \ni t \mapsto (x(t), \xi(t)) \in \Lambda_+$ be an integral curve of H_q and let us compute the length of the curve $t \mapsto x(t)$ for the metric $V(x) (dx)^2$. First we notice that

$$\frac{d}{dt} \psi(x(t)) =$$

$$\nabla \psi(x(t)) \cdot \dot{x}(t) = 2 \xi(t) \cdot \dot{x}(t) = 2 \|\xi(t)\| \cdot \|\dot{x}(t)\| = \sqrt{V(x(t))} \cdot \|\dot{x}(t)\|.$$

It is then clear that the length is

$$(3.10) \quad \int_a^b \sqrt{V(x(t))} \|\dot{x}(t)\| dt = \psi(x(b)) - \psi(x(a)).$$

In particular, $\psi(x(t)) \rightarrow 0$ when $t \rightarrow -\infty$ and after restricting Ω we may assume that $\Omega_y \subset \Omega_{y'}$, for all $0 < y < y' \leq y_0$ if $\Omega_y = \{x \in \Omega; \psi(x) < y\}$ and also that $\Omega = \Omega_{y_0}$. Then:

Proposition 3.1. Under the assumptions above we have $\psi(x) = d(x, 0)$ for all $x \in \Omega$.

Proof: Let $x \in \Omega$. We first show that $d(x, 0) \geq \psi(x) \stackrel{\text{def}}{=} y$. Let $\gamma: [a, b] \ni t \mapsto x(t) \in M$ be a C^1 curve with $x(a) = 0, x(b) = x$. Possibly after restriction to a segment of this curve we will have $x(t) \in \Omega_y$ for $a \leq t \leq b$ and not necessarily $x(b) = x$ but at least $\psi(x(b)) = y$. Then

$$y = \int_a^b \frac{d}{dt} \psi(x(t)) dt = \int_a^b \psi'(x(t)) \cdot \dot{x}(t) dt$$

so

$$y \leq \int_a^b \|\psi'(x(t))\| \|\dot{x}(t)\| dt = \int_a^b \sqrt{V} \|\dot{x}\| dt = \text{length of } \gamma.$$

Hence $y \leq d(x, 0)$.

On the other hand if $]-\infty, 0] \ni t \mapsto (x(t), \xi(t)) \in \Lambda_+$ is the bicharacteristic of q from $(0,0)$ to $(x, \psi'(x))$ then we have already seen that its projection in Ω_{y_0} is of length y and we can easily approach this curve by a sequence of curves $\gamma_j: [-T_j, 0] \rightarrow \Omega$, with $\gamma_j(-T_j) = 0$, $\gamma_j(0) = x$ and such that the length of γ_j tends to y .

#

In order to define a suitable Fourier integral operator, we first choose local coordinates by the Morse lemma so that $\psi(x) = \frac{1}{2} x^2$ and $V''(0)$ is diagonal. Then we can introduce the complex Lagrangian manifolds $L_\pm : \xi = \pm i \psi'(x)$ or rather $\xi = \pm i x$. Consider the phase function $f(x, y) = \frac{i}{2} (x-y)^2 - \frac{i}{4} x^2$ and its associated complex canonical transformation $\mathcal{H}_f : (y, i(x-y)) \mapsto (x, i(x-y) - \frac{i}{2} x)$. Then $\mathcal{H}_f(L_+)$ is given by $\xi = 0$ and $\mathcal{H}_f(L_-)$ is given by $x = 0$. Also $\mathcal{H}_f(\mathbb{R}^{2n})$ is the I-Lagrangian manifold given by $\xi = \frac{2}{i} \frac{\partial \phi}{\partial x}$ where $\phi(x) = \frac{1}{4} |x|^2$. (This terminology is due to Schapira and we refer to [24] where this notion is used and where also FBI transforms are studied). We define the FBI-transform (with $h = 1/\lambda$, see [24], section 7)

$$(3.11) \quad T_1 u(x, \lambda) = \lambda^{n/2} \int_{C_n} e^{i\lambda f(x, y)} u(y) dy$$

which maps $\mathcal{V}'(\mathbb{R}^n)$ into the space $H_\phi^{loc}(\mathbb{C}^n)$ of functions $v(x, \lambda)$ on $\mathbb{C}^n \times [1, \infty]$ holomorphic in x and such that for each compact set $K \subset V$ and

each $\epsilon > 0$, we have $\sup_K |v(x, \lambda)| e^{-\lambda \phi(x)} = e^{\lambda \epsilon}$. Here $C_n > 0$ is a suitable non-vanishing constant that will make T_1 (formally) unitary. If $u(y, \lambda)$ is an analytic symbol defined near 0 (i.e. $u \in H_0^{\text{loc}}(U)$, where $U \subset \mathbb{C}^n$ is an open neighborhood of 0) and we put $\tilde{u}(y, \lambda) = u(y, \lambda) e^{-\lambda \psi(y)}$, then $T_1 \tilde{u}(x, \lambda)$ can be defined by restricting the integration to a neighborhood of $0 \in \mathbb{R}^n$ and the function $T_1 \tilde{u}(x, \lambda)$ is then independent of the choice of neighborhood up to a term which is uniformly exponentially decreasing in a neighborhood of $0 \in \mathbb{C}^n$ (see [24] section 3.4). By a shift in the integration contour we see that $T_1 \tilde{u}(x, \lambda) \in H_0^{\text{loc}}(V)$ if V is a sufficiently small neighborhood of 0. This has been proved in a slightly more general setting in [24], section 4, and in the terminology of [24] we know that up to equivalence (modulo exponentially small terms) the mapping $H_{0,0} \ni u \rightarrow T_1(u e^{-\lambda \psi}) \in H_{0,0}$ is well defined and bijective ([24], Theorem 4.5, here $H_{0,0}$ denotes germs of analytic symbols at 0 (see [24] section 1)). We also recall from [24], that if u is a classical analytic symbol, then by stationary phase the same is true for $T_1(u e^{-\lambda \psi})$. (We recall the terminology: If $\Omega \subset \mathbb{C}^n$, then a formal classical analytic symbol of order m in Ω is a formal expression of the form $\sum_{j \in \mathbb{N}} \lambda^{m-j} u_j(x)$, where u_j are holomorphic and for every compact set $K \subset \Omega$ there is a constant $C > 0$ such that $|u_j(x)| \leq C^{j+1} (j!)$ in K . If $x_0 \in K$ then we get a germ of analytic symbol $u \in H_{0,x_0}$, well defined up to equivalence by

$$u(x, \lambda) = \sum_{0 \leq j \leq 1/\epsilon C} \lambda^{m-j} u_j(x), \quad x \in \text{neighborhood of } x_0$$

u is then called the realization of the formal analytic symbol. Sometimes, without any change in the terminology we also consider asymptotic sums for $j \in \frac{1}{2} \mathbb{N}$.

We shall also need to consider formal C^∞ symbols: For $m \in \mathbb{R}$, $k \in \mathbb{N}$, we define $S^{m,k}$ and $S_{1/2}^{m,k}$ to be the spaces of formal symbols $\sum_0^\infty u_j(x) \lambda^{m-j}$, where u_j are formal power-series in x with $u_j = \mathcal{O}(|x|^{(k-2j)_+})$ and where the sum ranges over $\mathbb{N}$ and $\frac{1}{2} \mathbb{N}$ respectively. We notice that $S^{m,k} \subset S^{m',k'}$ (and similarly in the half order case) if $m' - m \in \mathbb{N}$ (respectively $\frac{1}{2} \mathbb{N}$) and $m - k/2 \leq m' - k'/2$. If $u \in S^{m,k}$ we say that the total order of u is $\leq m - k/2$. We shall only consider the cases when $m \in \frac{1}{2} \mathbb{N}$. Let $S = \bigcup_{1/2 \mathbb{N}} S^{m,0}$,

$S_{1/2} = \bigcup_{1/2 \mathbb{N}} S_{1/2}^{m,0}$. Now $u \rightarrow T_1(u e^{-\lambda \psi})$ is a bijection in the space of germs of formal classical analytic symbols and examining the stationary phase development of $T_1(u e^{-\lambda \psi})$, when u is a formal classical analytic symbol of order m , we see that $T_1(u e^{-\lambda \psi}) \in S^{m,k} \iff u \in S^{m,k}$. On the other hand if $v \in S^{m,k}$ then for every $N \in \mathbb{N}$, we can find a formal classical analytic symbol u_N of order m such that $v \equiv u_N \pmod{S^{m-N,k+N}}$ and it is then clear how to define $T_1(v e^{-\lambda \psi}) \in S^{m,k}$. T_1 is a bijection $e^{-\lambda \psi} S^{m,k} \rightarrow S^{m,k}$ and also $e^{-\lambda \psi} S_{1/2}^{m,k} \rightarrow S_{1/2}^{m,k}$. Let $S^m(\mathbb{R}_+)$ be the space of formal classical symbols of order m : $\sum_{j \in \mathbb{N}} a_j \lambda^{m-j}$, ($a_j \in \mathbb{C}$) and put $S_{1/2}^m(\mathbb{R}_+) = S^m(\mathbb{R}_+) + S^{m-1/2}(\mathbb{R}_+)$. If $v_j \in S^{m_j, k_j}$, $j = 1, 2$, and $L(dx)$ denotes the Lebesgue measure on $\mathbb{C}^n$, then the scalar product

$$(3.12) \quad (v_1 | v_2)_\phi = \lambda^n \int v_1(x, \lambda) \overline{v_2(x, \lambda)} e^{-2\lambda \phi(x)} L(dx),$$

can be defined as a formal symbol by a formal stationary phase expansion. If $k_1 + k_2$ is even we get $(v_1 | v_2)_\phi \in S^{m_1+m_2-(k_1+k_2)/2}(\mathbb{R}_+)$ and if $k_1 + k_2$ is odd, then $(v_1 | v_2)_\phi \in S^{m_1+m_2-(k_1+k_2+1)/2}(\mathbb{R}_+)$. We also have the analogous results in the half degree case.

If C_n is conveniently chosen in (3.11) then for all $u_j \in S$ (or $S_{1/2}$) we have

$$(3.13) \quad (T_1(u_1 e^{-\lambda \psi}) | T_1(u_2 e^{-\lambda \psi}))_\phi = \lambda^{n/2} \int u_1(y, \lambda) \overline{u_2(y, \lambda)} e^{-2\lambda \psi(y)} dy,$$

where the right hand side also is defined by the corresponding stationary phase expansion. In proving (3.13) it suffices to treat the case when u_1, u_2 are formal classical analytic symbols, so that the two sides of (3.13) are well defined modulo $\mathcal{O}(e^{-\epsilon \lambda})$, by choosing realizations of u_1, u_2 and suitable integration cycles. Here we give only the formal verification, since by [24], section 4, it is clear that all integrals have suitable integration cycles and that the computations below are then valid with exponentially small errors: We notice that

$$\phi(x) = \sup_{t \in \mathbb{R}^n} (-\text{Im } f(x, t))$$

or

$$\phi(x) = \frac{i}{2} \text{v.c.}_t (F(x, t) - \overline{f(x, \bar{t})})$$

where v.c._t means "critical value with respect to (the complex variables) t ". We put $g(x, t) = -\overline{f(x, \bar{t})}$ and

$$(3.14) \quad \frac{2}{i} \psi(x, y) = \text{v.c.}_t (f(x, t) + g(y, t))$$

which is well defined since the leading homogeneous part of the polynomial $t \rightarrow f(x, t) + g(y, t)$ is a non-degenerate quadratic form. We notice that ψ is the unique entire function on $\mathbb{C}^{2n}$ which satisfies $\psi(x, \bar{x}) = \phi(x)$. Since $\det \frac{\partial^2 f}{\partial x \partial \bar{x}} \neq 0$ the relation (3.14) can be inverted:

$$(3.14)' \quad -g(y, t) = \text{v.c.}_x (F(x, t) - \frac{2}{i} \psi(x, y))$$

where the unique critical point is necessarily non-degenerate. (See [24] section 4). With $\tilde{u}_j = u_j e^{-\lambda \psi}$ we get

$$\begin{aligned} (T_1(\tilde{u}_1) | T_1(\tilde{u}_2))_\phi &= \\ &= \lambda^{2n} |C_n|^2 \int \int \tilde{u}_2(x, \lambda) e^{i\lambda(-f(z, \bar{x}) + f(z, y) - \frac{2}{i} \psi(z, \bar{z}))} \tilde{u}_1(y, \lambda) dy \, L(dz) dx \\ &= \lambda^{2n} |C_n|^2 (2i)^{-n} \int \int \int \overline{\tilde{u}_2(\bar{x}, \lambda)} e^{i\lambda(g(w, x) + f(z, y) - \frac{2}{i} \psi(z, w))} \tilde{u}_1(y, \lambda) dy \, dz \, dw \, dx \end{aligned}$$

Using (3.14)' and stationary phase we can eliminate the dz integral and get

$$\begin{aligned} (T_1(\tilde{u}_1) | T_1(\tilde{u}_2))_\phi &= \\ &= \mathcal{J}_1 \lambda^{3n/2} |C_n|^2 \int \int \overline{\tilde{u}_2(\bar{x}, \lambda)} e^{i\lambda(g(w, x) - g(w, y))} \tilde{u}_1(y, \lambda) dy \, dw \, dx. \end{aligned}$$

By the Kuranishi trick the $dy \, dw$ integral is just a constant times the identity operator acting on u_1 so

$$(T_1 \tilde{u}_1 | T_1 \tilde{u}_2)_\phi = \mathcal{J}_2 \lambda^{n/2} |C_n|^2 \int \overline{\tilde{u}_2(\bar{x}, \lambda)} \tilde{u}_1(x, \lambda) dx$$

Choosing $u_1 = u_2 = 1$ it is clear that $\mathcal{J}_2$ (which could be computed!) is > 0 and if we choose $C_n = \mathcal{J}_2^{-1/2}$ we get (3.13).

We now recall that $y_1, \dots, y_n$ were local coordinates on a Riemannian manifold. If the volume element is $a(y)dy$, $a > 0$, we put

$$(3.15) \quad T u = T_1(a^{1/2} u)$$

so that (3.13) becomes

$$(3.13)' \quad (T(u_1 e^{-\lambda \psi}) | T(u_2 e^{-\lambda \psi})) = \lambda^{n/2} \int u_1 \overline{u_2} e^{-2\lambda \psi} a \, dy$$

In the analytic case we showed in [24] that there exists a formal classical analytic symbol

$$\tilde{p}(x, \xi, \lambda) = \sum_0^\infty \tilde{p}_j(x, \xi) \lambda^{-j}$$

defined in a neighborhood of $(0, 0)$, such that if we pick a realization (i.e. a partial sum over $0 \leq j \leq \lambda/\epsilon C$), also denoted by $\tilde{p}$, and define the corresponding pseudo-differential operator $\tilde{p}(x, D_x, \lambda) = \tilde{p}(x, \frac{1}{\lambda} D_x, \lambda)$ by means of suitable contour integrals, then

$$(3.16) \quad \tilde{p} T \equiv T P$$

in the sense that $(\tilde{p} T u - T P u) e^{-\phi(x)/h} = \mathcal{O}(e^{-\epsilon_0/h})$ uniformly in a fixed neighborhood of 0 in $\mathbb{C}^n$ and with a constant $\epsilon_0 > 0$ independent of the choice of distribution u with support in a small fixed neighborhood of 0. The principal symbol $\tilde{p}$ of P is given by

$$(3.17) \quad \tilde{p} \circ \mathcal{H}_F = p$$

Since p vanishes on L , $\tilde{p}$ will vanish on $x=0$ and $\xi=0$:

$$(3.18) \quad \tilde{p}(x, \xi) = i \sum_{|\alpha| = |\beta| = 1} x^\alpha a_{\alpha, \beta}(x, \xi) \xi^\beta$$

Using the symplectic invariance of the fundamental matrix and the fact that $V'(0)$ is diagonalized in the y -coordinates, we see that

$$(3.19) \quad \sum x^\alpha a_{\alpha, \beta}(0,0) \xi^\beta = \sum_1^n \mu_j x_j \xi_j,$$

where $\mu_j = 2\sqrt{\lambda_j}$. We can also define $\tilde{P}: S^{m,k} \rightarrow S^{m-1,k}$, $S_{1/2}^{m,k} \rightarrow S_{1/2}^{m-1,k}$ by stationary phase expansions, and it is not difficult to see that

$$(3.20) \quad \tilde{P}(Tu) = TPu, \quad \forall u \in e^{-\lambda \Psi} S_{1/2}.$$

In the C^∞ case the $\tilde{p}_j$ are only defined as formal power series, but (3.17)-(3.20) remain valid. The self adjointness of P and (3.15)' imply that $\tilde{P}$ is formally self adjoint for the scalar product (3.12). We conclude this preliminary discussion by noting that the symplectic invariance of the subprincipal symbol implies that:

$$(3.21) \quad \tilde{P}_{-1}(0,0) = \frac{1}{2} \sum_1^n \mu_j.$$

We shall now study $Q = \lambda \tilde{P}$ on the spaces $S^{m,k}$ and $S_{1/2}^{m,k}$. We can write

$$Q = \sum_{|\alpha|=|\beta|=1} x^\alpha a_{\alpha\beta}(x, \tilde{D}_x) \partial_x^\beta + a(x, \tilde{D}_x, \lambda)$$

where a is a formal classical pseudo-differential operator of order 0.

We put

$$Q_0 = \sum_{|\alpha|=|\beta|=1} x^\alpha a_{\alpha\beta}(0,0) \partial_x^\beta + a_0(0,0)$$

Q_0 also conserves the spaces $S^{m,k}$ and $S_{1/2}^{m,k}$ and is self-adjoint for the scalar product (3.14). If we let $S_h^{m,k}$ denote the space of "homogeneous" elements $\lambda^m u_m(x)$ where u_m is a homogeneous polynomial of degree k , then $Q_0: S_h^{m,k} \rightarrow S_h^{m,k}$. We notice that $Q - Q_0$ is regularizing of degree $1/2$:

$Q - Q_0: S^{m,k} \rightarrow S^{m,k+1}$ and similarly in the half degree case.

The action of Q_0 on the polynomials is of course very easy to analyze. The monomials are eigenfunctions and the eigenvalue corresponding to $u = x^\alpha$ is $\sum (\alpha_j + \frac{1}{2}) \mu_j$. Let E_0 be such an eigenvalue. Then we can find a basis $e_1, \dots, e_{N_0}$ of eigenvectors with $e_j \in S_h^{m_j, k_j}$, $m_j - k_j/2 = 0$. More precisely, if $u \in S_{1/2}^{m,k}$, $(Q_0 - E_0)u = 0$, then there exist unique $\lambda_j \in S_{1/2}^{m-k/2}$ such

that $u = \sum_1^{N_0} \lambda_j e_j$. When $u \in S^{m,k}$ we have $\lambda_j \in S_{1/2}^{m-k/2+p_j}$, where $p_j = 0$ when $m_j - \frac{k}{2} \in \mathbb{Z}$ and $p_j = -1/2$ otherwise. If $E \notin \{ \sum_1^n (\alpha_j + \frac{1}{2}) \mu_j; \alpha \in \mathbb{N}^n \} = \text{Sp}(Q_0)$ then $Q_0 - E$ is invertible in the space of all polynomials and hence in the spaces $S^{m,k}$ and $S_{1/2}^{m,k}$. In particular, if γ is a simple oriented loop with E_0 in its interior and $\text{Sp}(Q_0) \cap \gamma$ in its exterior then

$$(3.22) \quad \Pi_0 = \oint_\gamma (E - Q_0)^{-1} dE, \quad \left(\oint_\gamma = \frac{1}{2\pi i} \int_\gamma \right),$$

is a projection $S^{m,k} \rightarrow S^{m,k} \cap \text{Ker}(Q_0 - E_0)$ (and idem in the $\frac{1}{2}$ degree case). If $E \notin \text{Sp}(Q_0)$ then $(Q - E)^{-1}$ is also well defined $S^{m,k} \rightarrow S^{m,k}$, $S_{1/2}^{m,k} \rightarrow S_{1/2}^{m,k}$. In fact, since $\Delta Q = Q_0 - Q: S_{(1/2)}^{m,k} \rightarrow S_{(1/2)}^{m,k+1}$ the Neumann series $(Q - E)^{-1} = (1 - (Q_0 - E)^{-1} \Delta Q)^{-1} (Q_0 - E)^{-1} = (Q_0 - E)^{-1} + (Q_0 - E)^{-1} \Delta Q (Q_0 - E)^{-1} + (Q_0 - E)^{-1} \Delta Q (Q_0 - E)^{-1} \Delta Q (Q_0 - E)^{-1} + \dots$ is defined asymptotically. Also

$$(Q - E)^{-1} - (Q_0 - E)^{-1}: S_{(1/2)}^{m,k} \rightarrow S_{(1/2)}^{m,k+1}.$$

Put

$$(3.23) \quad \Pi = \oint_\gamma (E - Q)^{-1} dE: S_{(1/2)}^{m,k} \rightarrow S_{(1/2)}^{m,k}.$$

It is clear that $[E, Q] = 0$ and that $\Pi - \Pi_0: S_{(1/2)}^{m,k} \rightarrow S_{(1/2)}^{m,k+1}$.

Lemma 3.2. *If $F \notin \text{Sp } Q_0$ is outside γ , then*

$$(3.24) \quad (F - Q)^{-1} \Pi = \oint_\gamma (F - E)^{-1} (E - Q)^{-1} dE.$$

Proof: If we let A be the right hand side of (3.24), then

$$(F - Q)A = \oint_\gamma (E - Q)(F - E)^{-1}(E - Q)^{-1} + (E - Q)^{-1} dE = 0 + \Pi = \Pi.$$

Hence $A = (F - Q)^{-1} \Pi$.

Lemma 3.3. *Π is a projection: $\Pi^2 = \Pi$.*

#

Proof : Let Γ have the same properties as γ and contain γ in its interior. Then

$$\begin{aligned} \Pi^2 &= \oint_{\Gamma} (F - Q)^{-1} \Pi dF = \oint_{\Gamma} \oint_{\gamma} (F - E)^{-1} (E - Q)^{-1} dE dF = \\ &= \oint_{\gamma} \left(\oint_{\Gamma} (F - E)^{-1} dF \right) (E - Q)^{-1} dE = \Pi \end{aligned}$$

since $\oint_{\Gamma} (F - E)^{-1} dF = 1$.

#

Proposition 3.4. $\Pi(e_1), \dots, \Pi(e_{N_0})$ is a basis for $\Pi(S_{1/2})$. More precisely :

(a) If $u \in \Pi(S_{1/2})$ is of total degree $\leq m_0$, then there exist

$$\lambda_1, \dots, \lambda_{N_0} \in \mathbb{R}_+ \text{ such that } u = \sum_{j=1}^{N_0} \lambda_j \Pi(e_j).$$

(b) If $\sum_{j=1}^{N_0} \lambda_j (\Pi(e_j) + v_j) = 0$ with $\lambda_j \in \mathbb{R}_+$, $v_j \in S_{1/2}$, $v_j \neq 0$, then

$$\text{then } \lambda_1 = \dots = \lambda_{N_0} = 0.$$

Proof : (a) First we take $\lambda_j^0 \in S_{1/2}^{m_0}$ such that $\Pi_0 u = \sum_{j=1}^{N_0} \lambda_j^0 e_j$. Then $\Pi \Pi_0 u = \sum_{j=1}^{N_0} \lambda_j^0 \Pi(e_j)$. But $\Pi \Pi_0 u = u + (\Pi \Pi_0 - \Pi^2) u = u + \Pi(\Pi_0 - \Pi)u = u - u^{(1)}$, where $u^{(1)} \in \Pi(S_{1/2})$ is of degree $m_0 - 1/2$. Hence $u = \sum_{j=1}^{N_0} \lambda_j^0 \Pi(e_j) + u^{(1)}$ and by iteration we get $u^{(1)} = \sum_{j=1}^{N_0} \lambda_j^1 \Pi(e_j) + u^{(2)}$ where $\lambda_j^1 \in S_{1/2}^{m_0-1/2}$ and $u^{(2)}$ of degree $\leq m_0 - 1, \dots$. We take $\lambda_j = \lambda_j^0 + \lambda_j^1 + \dots$.

(b) : Suppose that $\lambda_j \in S_{1/2}^{m_0}(\mathbb{R}_+)$ and let λ_j^0 be the leading homogeneous parts of degree m_0 . Then we get $\sum_{j=1}^{N_0} \lambda_j^0 e_j = 0$ hence $\lambda_j^0 = 0$ so $\lambda_j \in S_{1/2}^{m_0-1/2}(\mathbb{R}_+) \dots$

#

After rearranging the $\Pi(e_j) \in S^{m_j, k_j}$ ($m_j - k_j/2 = 0$) we may assume that

(3.25) k_j is even for $1 \leq j \leq N_0$ and odd for $N_0 + 1 \leq j \leq N_0$.

Then $(\Pi(e_j) | \Pi(e_k)) \in S^0(\mathbb{R}_+)$ when j, k belong to the same group, otherwise $(\Pi(e_j) | \Pi(e_k)) \in S^{-1/2}(\mathbb{R}_+)$. Moreover we may assume that $e_1, \dots, e_{N_0}$ is an orthonormal system so that $(\Pi(e_j) | \Pi(e_k)) = \delta_{jk} + \text{symbol of lower order}$.

Then $((\Pi(e_j) | \Pi(e_k)))_{1 \leq j, k \leq N_0} = I + A$, where

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}, \quad A_{11}, A_{22} \in S^{-1}, \quad A_{12}, A_{21} \in S^{-1/2}.$$

By a binomial series we construct $(I + A)^{-1/2} = I + B$ where $B = B^*$ has the same structure. We put

$$\begin{pmatrix} f_1 \\ \vdots \\ f_{N_0} \end{pmatrix} = (I + B) \begin{pmatrix} \Pi(e_1) \\ \vdots \\ \Pi(e_{N_0}) \end{pmatrix}$$

Then

$$\left(\begin{pmatrix} f_1 \\ \vdots \\ f_{N_0} \end{pmatrix} \middle| \begin{pmatrix} f_1 \\ \vdots \\ f_{N_0} \end{pmatrix} \right) = (I + B)(I + A)(I + B) = I.$$

Proposition 3.4 remains valid for the system $f_1, \dots, f_{N_0}$ so we have an O.N. basis for $\Pi(S_{1/2})$.

Proposition 3.5. We have $f_j \in S^{\tilde{m}_j, \tilde{k}_j}$ with $\tilde{m}_j - \tilde{k}_j/2 = 0$ and $\tilde{k}_j$ is even for $1 \leq j \leq M_0$ and odd for $M_0 + 1 \leq j \leq N_0$.

Proof : Assuming for instance that $M_0 > 1$, then f_1 is the sum of two types of terms u_j and v_j , where $u_j \in S^{m_j, k_j}$ with $m_j - k_j/2 = 0$ and k_j is even and $v_j \in S^{m_j-1/2, k_j}$ with $m_j - k_j/2 = 0$ and k_j is odd. For v_j we also have $v_j \in S^{\bar{m}_j, \bar{k}_j}$, where $\bar{m}_j = m_j - 1/2$, $\bar{k}_j = k_j - 1$ (even) and $\bar{m}_j - \frac{1}{2}\bar{k}_j = 0$. Hence all our terms are actually of the classes S^{m_j, k_j} with $m_j = \frac{1}{2}k_j$ and k_j even so $f_1 \in S^{\tilde{m}_1, \tilde{k}_1}$ with $\tilde{m}_1 = \max(m_j)$, $\tilde{k}_1 = \max(k_j)$. The same argument works for a vector f_j in the second group.

#

We next remark that $Q f_k \in S_{\tilde{m}_k, \tilde{k}_k}$ and that $Q f_k \equiv E_0 f_k \pmod{S_{\tilde{m}_k, \tilde{k}_k+1}}$. Hence the matrix of $Q|_{\Pi(S)}$ with respect to the O.N. basis $f_1, \dots, f_{N_0}$ becomes

$$(3.26) \quad ((Q f_k | f_j)) = E_0 I + \begin{pmatrix} Q_{11} & Q_{12} \\ Q_{21} & Q_{22} \end{pmatrix},$$

where Q is a hermitian matrix and $Q_{11}, Q_{22} \in S^{-1}(\mathbb{R}_+)$, $Q_{12}, Q_{21} \in S^{-1/2}$.

Theorem 3.6. There exists (in the sense of Proposition 3.4) an orthonormal basis of eigenvectors of Q in $\Pi(S_{1/2})$; $g_j \in S_{\tilde{m}_j, \tilde{k}_j}^{1/2}$ with $\tilde{m}_j = \frac{1}{2} \tilde{k}_j$ and the corresponding eigenvalues of Q belong to $E_0 + S_{1/2}^{-1/2}(\mathbb{R}_+)$. In the case when $M_0 = 0$ or N_0 in (3.25) then we may choose $g_j \in S_{\tilde{m}_j, \tilde{k}_j}$ and the eigenvalues belong to $E_0 + S^{-1}(\mathbb{R}_+)$. Moreover the $\tilde{k}_j$ are odd when $M_0 = 0$ and even when $M_0 = N_0$.

Proof: It is no restriction to assume that $E_0 = 0$. All the computations will be made with respect to the basis $f_1, \dots, f_{N_0}$. We write $Q = \sum_{v=1/2}^{\infty} h^v Q_v$, where Q_v are hermitian scalar matrices. Then $Q_N = \sum_{1/2 < v \leq N} h^v Q_v$ is an analytic function of $h^{1/2}$ so the

eigenvalues of Q_N ; $\lambda_1^{(N)}, \dots, \lambda_{N_0}^{(N)}$ are analytic functions of $h^{1/2}$. Moreover, if $M > N$, then if we arrange the eigenvalues suitably, $\lambda_j^{(M)} - \lambda_j^{(N)} = \mathcal{O}(h^N)$, and hence we can define asymptotic eigenvalues $\lambda_j \in S_{1/2}^{-1/2}$. Notice that the λ_j are the roots of $\det(Q - E)$. Suppose that λ_1 has the multiplicity d in the obvious sense. Then there exists N_1 such that if $N \geq N_1$ then $\lambda_1^{(N)}, \dots, \lambda_d^{(N)}$ all differ by at most $\mathcal{O}(h^N)$, while $|\lambda_j^{(N)} - \lambda_k^{(N)}| \geq \text{const}|h|^{N_1-1}$ if $j \leq d, d+1 \leq k$. Let $E_1^{(N)}$ be the eigenspace of Q_N associated to $\lambda_1^{(N)}, \dots, \lambda_d^{(N)}$. The spectral projection $\Pi_{E_1^{(N)}}$ is a meromorphic function of $h^{1/2}$ near

the origin and $\|\Pi_{E_1^{(N)}}\| = 1$ for $h^{1/2}$ real so we conclude that $\Pi_{E_1^{(N)}} \in S_{1/2}^0$

for $N \geq N_1$. Representing $\Pi_{E_1^{(N)}}$ as a contour integral over $|E - \lambda_1^{(N)}| = |h|^{N_1}$

for $N \geq N_1$, we also see that $\Pi_{E_1^{(N)}} = \Pi_{E_1^{(N_1)}} + \mathcal{O}(|h|^{N-N_1})$, for $N > N > N_1$. Let $g_1^{(N_1)}, \dots, g_d^{(N_1)}$ be an O.N. basis in $E_1^{(N_1)}$ and define $g_1^{(N)}, \dots, g_d^{(N)} \in E_1^{(N)}$

as the orthonormalization of $\Pi_{E_1^{(N)}}(g_1^{(N_1)}), \dots, \Pi_{E_1^{(N)}}(g_d^{(N_1)})$. Then it is easy to see that the sequence $g_j^{(N)}$ defines an element $g_j \in S_{1/2}^0$, and that g_j form an orthonormal system in $\text{Ker}(Q - \lambda_1)$. Doing the same thing for each (multiple) eigenvalue we get the desired basis. The second part of the theorem is obtained by the same arguments starting from the fact that this time $Q \in S^{-1}(\mathbb{R}_+)$.

We finally get the main result of this section for the operator P :

Theorem 3.7. Let E_0 be as above. Then there exist $a_j \in S_{\tilde{m}_j, \tilde{k}_j}^{1/2}$, $1 \leq j \leq N_0$ with $\tilde{m}_j = \frac{1}{2} \tilde{k}_j$ such that the functions $u_j = a_j e^{-\lambda_j \psi}$ form an orthonormal system for the formal inner product $\lambda^{n/2} (u|v)_{L^2}$ (cf. 3.15') and such that $\lambda P u_j = E_j u_j$ where $E_j \in E_0 + S_{1/2}^{-1/2}(\mathbb{R}_+)$. In the case when $M_0 \in \{0, N_0\}$ certain half-degree terms disappear exactly as in Theorem 3.6.

Remark 3.8. : One could avoid using FBI transforms in the proof of Theorem 3.7, and instead work with $Q = \lambda e^{\lambda \psi} P e^{-\lambda \psi}$ for instance in the coordinates y where $\psi = y^2/2$. Then Q_0 will have a somewhat less pleasant form and its eigenfunctions will be Hermite polynomials rather than homogeneous polynomials. (See the formal expansions in Damburg-Propin [28]).

Remark 3.9. : The sufficient conditions for the disappearance of the $1/2$ degree terms in the eigenvalues are equivalent to the condition that for all $\alpha \in \mathbb{N}^n$ such that $\sum (\alpha_j + \frac{1}{2}) \mu_j = E_0$ the numbers $\sum_1^n \alpha_j$ have the same parity.

Remark 3.10. : If we write $a_j = \sum_0^\infty a_{j, \tilde{m}_j - v}(x) \lambda^{\tilde{m}_j - v}$ and define the formal classical symbol in a neighborhood Ω of 0:

$$\tilde{a}_j(x, \lambda) = \sum_0^\infty \tilde{a}_{j, \tilde{m}_j - v}(x) \lambda^{-v},$$

where $\tilde{a}_{j, k}$ is a C^∞ function with $a_{j, k}$ as its Taylor series at 0, then

$(\lambda P - E_j) \hat{u}_j = b(x, \lambda) e^{-\lambda \psi(x)}$, where $\hat{u}_j = \hat{a}_j e^{-\lambda \psi}$ and $b(x, \lambda) = \sum_{\nu=0}^{\infty} \hat{b}_{j, \hat{m}_j - \nu}(x) \lambda^{\hat{m}_j - \nu}$ has the property that each b_ν vanishes to infinite order at $x=0$. If Ω is sufficiently small and has a suitable shape, we can then modify the $\hat{a}_{j, \hat{m}_j - \nu}(x)$ by terms vanishing to infinite order so that the b_ν vanish identically in Ω . Indeed, this amounts to solve repeatedly transport equations of the form

$$(v(x, \frac{\partial}{\partial x}) - f(x))u(x) = v(x),$$

where $v = \mathcal{O}(|x|^\omega)$ and v is a fixed real vector field vanishing only at 0 (equal to $\sum \frac{\partial \psi}{\partial x_j} \frac{\partial}{\partial x_j}$ in the case $M = \mathbb{R}^n$) and whose linearisation at 0 has only strictly positive eigenvalues. Then for arbitrarily small suitable Ω we know that $x \in \Omega \Rightarrow x_t = \exp t v(x) \in \Omega$, $t \leq 0$ and that

$$| \exp t v(x) | \leq C_\alpha e^{-|t|/C} |x|, \quad (C > 0).$$

This is easy to prove directly and we refer to B. Lascar - Sjöstrand [15] for more elaborate constructions of this type. It is then clear that we have a C^∞ solution u with $u = \mathcal{O}(|x|^\omega)$ of the form

$$(3.27) \quad u(x) = \int_{-\infty}^0 v(x_t) \exp \left(\int_t^0 f(x_s) ds \right) dt.$$

Summing up, if Ω is a small neighborhood of 0 then we can find formal classical symbols $\hat{a}_j(x, \lambda) \sim \sum_{\nu \in \frac{1}{2}\mathbb{N}} \hat{a}_{j, \hat{m}_j - \nu}(x) \lambda^{\hat{m}_j - \nu}$ in Ω representing the

a_j in Theorem 3.7, such that $(\lambda P - E_j)(a_j e^{-\lambda \psi}) = 0$ in Ω . If we let $\hat{a}_j$ be an asymptotic sum of $\hat{a}_j$ and $\hat{E}_j$ an asymptotic sum of $E_j(\lambda)$, then

$$(3.28) \quad (\lambda P - \hat{E}_j)(\hat{a}_j e^{-\lambda \psi}) = \hat{b}(x, \lambda) e^{-\lambda \psi}, \quad \hat{b} \in S^{-\infty}(\Omega \times \mathbb{R}_+).$$

Remark 3.11. If $\mathcal{J}u(y) = \bar{u}(y)$ for functions of real variables, then we see that $\mathcal{J}$ commutes with T if we define $(\mathcal{J}v)(x) = \bar{v}(\bar{x})$ when v is a function of complex variables. Moreover $\mathcal{J}$ commutes with P so $\mathcal{J}$ commutes with the transformed operator as well. The functions $e_1, \dots, e_{N_0}$ can be taken as monomials with a real coefficient and are hence real ($\mathcal{J}e_j = e_j$). It is

easy to check that Π commutes with $\mathcal{J}$, so $\Pi(e_j)$ and $(\Pi(e_j), \Pi(e_k))$ are real valued, hence the matrices A, B are real, $f_1, \dots, f_{N_0}$ are real and the matrix Q is real. We can then choose g_j in Theorem 3.6 real valued and hence the same for $a_j, u_j, \hat{a}_j, \hat{u}_j$ in Theorem 3.7 and Remark 3.10.

We end this section with an example where $1/2$ degrees appear in the asymptotic eigenvalue. To simplify the computations we work on the FBI-transform side but the same arguments can be applied directly to the Schrödinger operators.

Consider

$$(3.28) \quad Q = 2x_1 \partial_{x_1} + x_2 \partial_{x_2} + x_2(ax_2 + \bar{a} \partial_{x_2}) \partial_{x_1} + x_1(ax_2 + \bar{a} \partial_{x_2}) \partial_{x_2}.$$

We assume that a change of variables has already taken place so that $\phi = \frac{1}{2} |x|^2$ (instead of $\frac{1}{4} |x|^2$). The Weyl symbol of Q restricted to Λ_ϕ is real, so Q is self adjoint with respect to the scalar product (3.14). Moreover the principal symbol $i(2x_1 \xi_1 + x_2 \xi_2)$ behaves like $|x|^2$ on Λ_ϕ , and this is the degenerate ellipticity that the FBI transforms of the Schrödinger operators under consideration should have. After the change of variables $x = \lambda^{-1/2} y$ we get

$$Q = Q_0 + \lambda^{-1/2} Q_1,$$

where

$$Q_0 = 2y_1 \partial_{y_1} + y_2 \partial_{y_2},$$

$$Q_1 = y_2(a y_2 + \bar{a} \partial_{y_2}) \partial_{y_1} + y_1(a y_2 + \bar{a} \partial_{y_2}) \partial_{y_2}.$$

The same change of variables transforms $S_{1/2}^{m,k}$ with $m = \frac{1}{2}k$ into the space of symbols $u = u_0 + \lambda^{-1/2} u_1 + \lambda^{-1} u_2 + \dots$, where $u_j(y)$ is a polynomial of degree $k+j$. Such a symbol is an eigenvector for Q with eigenvalue $E = E_0 + \lambda^{-1/2} E_1 + \lambda^{-1} E_2 + \dots$ if and only if we have the system

$$(0) \quad (Q_0 - E_0)u_0 = 0$$

$$(1) \quad (Q_0 - E_0)u_1 + (Q_1 - E_1)u_0 = 0$$

$$(2) \quad (Q_0 - E_0)u_2 + (Q_1 - E_1)u_1 = 0$$

We now work in the Bargmann - Fock space with scalar product

$$(3.29) \quad (u|v) = \int u(x) \overline{v(x)} e^{-2\Phi(x)} L(dx)$$

where the monomials, suitably normalized, form an O.N. basis. Let $\mathcal{E}_0$ be the eigenspace associated to the eigenvalue E_0 . Then (0) amounts to the condition that $u_0 \in \mathcal{E}_0$. If Π_0 is the orthogonal projection onto $\mathcal{E}_0$ then (1) has a solution iff $(\Pi_0 Q_1 - E_1)u_0 = 0$. In particular, if the self adjoint operator $\Pi_0 Q_1|_{\mathcal{E}_0}$ is non degenerate then $E_1 \neq 0$ and $\frac{1}{2}$ degrees do appear.

In our particular example let us take $E_0 = 2$. Then $\mathcal{E}_0$ is spanned by the orthogonal system y_1, y_2^2 . Then $Q_1(y_1) = a y_2^2$, $Q_1(y_2^2) = 2a y_2^2 y_1 + 2\bar{a} y_1$. Hence $\Pi_0 Q_1(y_2^2) = 2\bar{a} y_1$ and $\Pi_0 Q_1(y_1) = a y_2^2$ so the matrix of $\Pi_0 Q_1$ in the base y_1, y_2^2 is

$$\begin{pmatrix} 0 & 2\bar{a} \\ a & 0 \end{pmatrix}$$

which is non degenerate if we choose $a \neq 0$.

The same type of computations work directly on

$$h^2 D_{x_1}^2 + x_1^2 + 2(h^2 D_{x_2}^2 + x_2^2) + V(x)$$

where V is a homogeneous polynomial of degree 3. Putting $x_j = h^{1/2} y_j$, we get

$$h(D_{y_1}^2 + y_1^2 + 2(D_{y_2}^2 + y_2^2) + h^{1/2} V(y_1, y_2))$$

and $D_{y_1}^2 + y_1^2 + 2(D_{y_2}^2 + y_2^2)$ has the double eigenvalue $E_0 = 1$. Again we get $E_1 \neq 0$ if we take $V(x) = x_1^2 x_2$.

4. FURTHER INFORMATION IN THE ANALYTIC CASE

In this section we assume some familiarity with the machinery of analytic pseudodifferential operators and related notions. Actually, we shall

use the terminology and some results of [24]. Throughout this section we assume that M and V are real analytic and we keep the other assumptions from section 3. Then as we already pointed out, the distance function $\psi(x)$ is analytic near 0 and the symbol a in (3.11) is a classical analytic symbol of order 0. The transformed operator $Q = \lambda \tilde{P}$ is then a formal analytic classical pseudodifferential operator of order 1, whose symbol is defined in a complex neighborhood of $(0,0)$.

Consider the vector field

$$v = \sum_{|\alpha|=|\beta|=1} x^\alpha a_{\alpha,\beta}(x,0) \partial_x^\beta$$

As in Remark 3.10 if Ω is a suitable small open neighborhood of 0, then $x \in \Omega \Rightarrow x_t = \exp t v(x) \in \Omega$ and $|x_t| \leq C e^{-|t|/C} |x|$ for all $t \leq 0$, $x \in \Omega$. If $f(x)$ is a holomorphic function in Ω with $f(0) \notin \text{Sp}(Q_0)$, then for every v holomorphic in Ω , there is a unique holomorphic function u in Ω such that

$$(4.1) \quad (v - f)u = v \quad \text{in } \Omega$$

Moreover, if v vanishes to the order k at 0, the same is true for u . At the level of formal power series all this is easily verified and in particular we know that u is unique. To show the existence of a holomorphic solution u we can clearly reduce ourselves to the case when $v(x) = \mathcal{O}(|x|^N)$, where $N > 0$ can be chosen arbitrarily large. But then (3.27) gives the desired solution.

Let $\mathcal{S}^{m,k}(\Omega)$ ($\mathcal{S}_{1/2}^{m,k}(\Omega)$) be the space of formal symbols

$$(4.2) \quad u(x, \lambda) = \sum_{j \in \mathbb{N} \ (j \in \frac{1}{2} \mathbb{N})} u_j(x) \lambda^{m-j}$$

where u_j is holomorphic in Ω and vanishes to the order $(k - 2j)_+$ at the origin. Using the above observation about the operator $v - f$, which for a suitable f appears as a transport operator, it is not difficult to see that if we define $Q : \mathcal{S}^{m,k} \rightarrow \mathcal{S}^{m,k}$ ($\mathcal{S}_{1/2}^{m,k} \rightarrow \mathcal{S}_{1/2}^{m,k}$) by formal asymptotic expansion, then for $E \notin \text{Sp}(Q_0)$, $Q - E$ becomes a bijection. Let $\mathcal{S}_a^{m,k}(\Omega)$,

$\mathcal{S}_{1/2,a}^{m,k}$ be the subspaces of $\mathcal{S}^{m,k}$, $\mathcal{S}_{1/2}^{m,k}$ of symbols which are also formal analytic symbols. That is, if u is of the form (4.2) then we require in addition that for every $K \subset \subset \Omega$, there exist $C > 0$ such that $|u_j(x)| \leq C^{j+1} j^j$ for $x \in K$, $j = 0, 1, 2, \dots$. The main step in this section will be to prove the following.

Theorem 4.1. For $E \in Sp(Q_0)$, $Q - E$ is a bijection $\mathcal{S}_a^{m,k}(\Omega) \rightarrow \mathcal{S}_a^{m,k}(\Omega)$, and similarly in the half order case.

This theorem can certainly be proved using tricky quasi-norms and it is closely related to results of Metivier [18]. We have preferred to give a new type of proof, perhaps more abstract, but less technical. It is possible that it could be improved so as to become more direct.

Let ϕ be a strictly plurisubharmonic and real analytic function defined near the origin in $\mathbb{C}^n$ with $\phi(0) = 0$, $\phi'(0) = 0$. As we saw in [24], Remark 4.3, when defining pseudodifferential operators in the complex domain, we are not obliged to use the standard phase. Another very natural phase, that we shall use here (see also [24], p. 160), is given by $\frac{2}{\pi}(\psi(x, \theta) - \psi(y, \theta))$, where $\psi(x, y)$ is the unique holomorphic function defined near the origin in $\mathbb{C}^{2n}$, such that $\psi(x, \bar{x}) = \phi(x)$. Then if $a(x, \theta, \lambda)$ is an analytic symbol defined near the origin in $\mathbb{C}^{2n}$ we consider the pseudodifferential operator

$$(4.3) \quad Au(x, \lambda) = \left(\frac{\lambda}{2\pi i}\right)^n \iint e^{2\lambda(\psi(x, \theta) - \psi(y, \theta))} a(x, \theta, \lambda) u(y, \lambda) dy d\theta.$$

Naturally (4.3) is still only formal and we should specify the function space and the integration contour. Let Ω be a small strictly pseudoconvex neighborhood of 0 and let $H_\phi(\Omega)$ denote the space of functions $u(x, \lambda)$ on $\Omega \times \mathbb{R}_+$, holomorphic in Ω and such that $\|u(\cdot, \lambda)\|_{H_\phi} \leq C_\epsilon e^{\epsilon\lambda}$, $\lambda > 0$, for every $\epsilon > 0$. Here $\|\cdot\|_{H_\phi}$ is the L^2 norm over Ω , with respect to the density $e^{-2\lambda\phi} L(dx)$. We now choose the contour $\theta = \bar{y}$, $y \in \Omega$. Then A becomes a well defined operator $H_\phi(\Omega) \rightarrow H_\phi(\Omega)$:

$$(4.4) \quad Au(x, \lambda) = \left(\frac{\lambda}{\pi}\right)^n \int_\Omega e^{2\lambda\psi(x, \bar{y})} a(x, \bar{y}, \lambda) u(y, \lambda) e^{-2\lambda\phi(y)} L(dy).$$

The adjoint A^* is of the same form, with $a(x, \bar{y}, \lambda)$ replaced by $\overline{a(y, \bar{x}, \lambda)}$, so it is the realization of an operator of the form (4.3) with $a(x, \theta, \lambda)$ replaced by $a(\bar{\theta}, \bar{x}, \lambda)$.

So far λ has been a large positive parameter, but we shall need also complex values $\lambda = r e^{i\omega}$. Then we define H_ϕ as before, replacing $\mathbb{R}_+$ by $\mathbb{C}$ and λ by r in the definition of the density. When realizing (4.3) it is clear that we will have to modify the choice of contour. We specialize by choosing $\phi(x) = \frac{1}{2}|x|^2$. Then $\psi(x, y) = \frac{1}{2}xy$ and with this choice $a(x, i\xi, \lambda)$ is the symbol in the classical sense, of the operator (4.3). We realize (4.3) by choosing the contour $\theta = e^{-i\omega} \bar{y}$, where $\lambda = r e^{i\omega}$. Then the generalization of (4.4) becomes

$$(4.5) \quad Au(x, \lambda) = \left(\frac{r}{\pi}\right)^n \int e^{r x \bar{y}} a(x, e^{-i\omega} \bar{y}, \lambda) u(y, \lambda) e^{-r|y|^2} L(dy)$$

which again maps $H_\phi(\Omega)$ into itself. The adjoint is

$$(4.6) \quad A^*u(x, \lambda) = \left(\frac{r}{\pi}\right)^n \int e^{r x \bar{y}} \overline{a(y, e^{-i\omega} \bar{x}, \lambda)} u(y, \lambda) e^{-r|y|^2} L(dy).$$

Considering r as the large parameter, the symbol of A in the classical sense is $a(x, e^{-i\omega} \xi, i e^{i\omega} r)$ and that of A^* is $\overline{a(-i \bar{\xi}, e^{-i\omega} \bar{x}, e^{i\omega} r)}$. Expressed in terms of $A(x, \xi, \lambda) = a(x, i\xi, \lambda)$, these quantities become $A(x, e^{-i\omega} \xi, e^{i\omega} r)$ and $\overline{A(i \bar{\xi}, i e^{i\omega} x, e^{i\omega} r)}$ respectively.

When $A = Q$ we have

$$(4.7) \quad A(x, \xi, \lambda) = i \lambda \sum_{|\alpha|=|\beta|=1} x^\alpha a_{\alpha, \beta}(x, \xi) \xi^\beta + a(x, \xi, \lambda)$$

where a is of order 0. The symbols of A and A^* with r as the large parameter, are then

$$(4.8) \quad i r \sum_{|\alpha|=|\beta|=1} x^\alpha a_{\alpha, \beta}(x, e^{-i\omega} \xi) \xi^\beta + a(x, e^{-i\omega} \xi, e^{i\omega} r)$$

and

$$(4.9) \quad i r \sum_{|\alpha|=|\beta|=1} \overline{\xi^\alpha} a_{\alpha,\beta}(\overline{i\xi}, i e^{i\omega} x) x^\beta + a(\overline{i\xi}, i e^{i\omega} x, e^{i\omega} r) .$$

In view of (3.19) and the fact that $a_0(0,0) = \frac{1}{2} \sum_{j=1}^n \nu_j$, we see that the symbols (4.8), (4.9) agree with $A(x, \xi, r)$ modulo $\lambda \mathcal{O}(|(x, \xi)|^3) + \mathcal{O}(|(x, \xi)|)$, or in other words that the operator Q_0 is not only a good approximation for Q when $\lambda > 0$ but also for A and A^* in general. For simplicity we shall write Q and Q^* instead of A and A^* in (4.5), (4.6). Let $E \in \mathbb{C} \setminus \text{Sp}(Q_0)$ and put $P = Q - E$. Our estimates and results below will be uniform with respect to E , when E varies in a compact set. We are now essentially in a situation where [25], Theorem 2.1 and Remark 2.3 apply (using Q_0 as a "localized" operator) and combining (2.20) of [25] with simple estimates in the elliptic region, we get

Proposition 4.2. If $\Omega_0 \subset \subset \Omega$ is an open neighborhood of 0 then there is a constant $C > 0$ such that

$$(4.10) \quad \|v\|_{H^\Phi(\Omega_0)} \leq C(\|P^*v\|_{H^\Phi(\Omega)} + \|v\|_{H^\Phi(\Omega \setminus \Omega_0)})$$

for every $v \in H^\Phi(\Omega)$ and for every $|\lambda| \geq 1$.

It should be noticed also that

$$(4.11) \quad \|P^*v\|_{H^\Phi(\Omega)} \leq \text{const.} |\lambda| \|v\|_{H^\Phi(\Omega)}, \quad \forall v \in H^\Phi(\Omega),$$

and similarly for $\nabla_\lambda P^*$.

It follows that the operator

$$\mathcal{A}: H^\Phi(\Omega) \ni v \rightarrow (P^*v, v|_{\Omega \setminus \Omega_0}) \in H^\Phi(\Omega) \times H^\Phi(\Omega \setminus \Omega_0)$$

is injective with closed image. Let $\mathcal{B} = (\mathcal{A}^* \mathcal{A})^{-1} \mathcal{A}^*$ be its "optimal" left inverse. Then $\|\mathcal{B}\| = \mathcal{O}(|\lambda|)$ for $|\lambda| \geq 1$ and since

$$\nabla_\lambda \mathcal{B} = (\mathcal{A}^* \mathcal{A})^{-1} ((\nabla_\lambda \mathcal{A}^*) \mathcal{A} + \mathcal{A}^* (\nabla_\lambda \mathcal{A})) (\mathcal{A}^* \mathcal{A})^{-1} \mathcal{A} + (\mathcal{A}^* \mathcal{A})^{-1} \nabla_\lambda \mathcal{A}^*$$

we find $\|\nabla_\lambda \mathcal{B}\| = \mathcal{O}(|\lambda|^3)$. Here when estimating the gradient of the adjoint, we notice that when $u, v \in L^2(\Omega)$ and $u, v \in L^2(\Omega) \cap \mathcal{H}^1(\Omega)$ respectively, then

$$\partial_r(u|v)_{L_\Phi^2} = (Fu|v)_{L_\Phi^2}, \quad \partial_r(u|v)_{H_\Phi} = (Fu|v)_{H_\Phi}, \text{ where}$$

$$Fu = -2\Phi u \quad \text{and} \quad Fu = B_{\Phi, \Omega}(-2\Phi)B_{\Phi, \Omega} u \text{ respectively.}$$

Here $B_{\Phi, \Omega}$ is the orthogonal projection $L_\Phi^2(\Omega) \rightarrow H_\Phi(\Omega)$. Thus in both cases $\|F\| = \mathcal{O}(1)$, $|\lambda| \rightarrow \infty$. If $A = A_\lambda$ is a C^1 -family of bounded operators between spaces of the type $L^2(\Omega)$ or $L^2(\Omega) \cap \mathcal{H}^1(\Omega)$, $\lambda = r e^{i\omega}$, then if A^* denotes the adjoint with respect to the r -dependent scalar products, we get

$$\partial_r A^* = (FA - AF + \partial_r A)^*, \quad \partial_\omega A^* = (\partial_\omega A)^*,$$

so

$$\|\nabla_\lambda A^*\| = \mathcal{O}(1)(\|A\| + \|\nabla_\lambda A\|).$$

Since $B_{\Phi, \Omega}$ is the adjoint of the inclusion $L^2(\Omega) \cap \mathcal{H}^1(\Omega) \rightarrow L^2(\Omega)$ we get $\|\nabla_\lambda B_{\Phi, \Omega}\| = \mathcal{O}(1)$.

The adjoint of $\mathcal{A}$ is

$$\mathcal{A}^*: H_\Phi(\Omega) \times H_\Phi(\Omega \setminus \Omega_0) \ni (u, \tilde{u}) \rightarrow Pu + B_{\Phi, \Omega}(1_{\Omega \setminus \Omega_0} \tilde{u}) \in H_\Phi(\Omega),$$

we also introduce,

$$\mathcal{B}^*: H_\Phi(\Omega) \rightarrow H_\Phi(\Omega) \times H_\Phi(\Omega \setminus \Omega_0) \\ w \rightarrow (Cw, Dw)$$

whose norm is $\mathcal{O}(|\lambda|)$ and satisfies $\|\nabla_\lambda \mathcal{B}^*\| = \mathcal{O}(|\lambda|^3)$, when $|\lambda| \geq 1$. Since $\mathcal{B}\mathcal{A} = I$ we have $\mathcal{A}^*\mathcal{B}^* = I$ which we write as

$$(4.12) \quad I = PC + B_{\Phi, \Omega}^{-1} 1_{\Omega \setminus \Omega_0} D.$$

Hence for $v \in H^\Phi(\Omega)$

$$(4.13) \quad PCv = v - B_{\phi, \Omega} (1_{\Omega \setminus \Omega_0} Dv)$$

We next need to have an idea about $B_{\phi, \Omega}$. The following discussion is valid for general real analytic strictly plurisubharmonic weight functions (possibly after decreasing Ω). Let $f \in L^2_{\phi}(\Omega)$, $g = Bf \in H_{\phi}(\Omega)$ ($B = B_{\phi, \Omega}$). Then $\|g\|_{H_{\phi}(\Omega)} \leq \|f\|_{L^2_{\phi}(\Omega)}$. As we saw in [15], [24], we can realize the identity operator on $H_{\phi}(\Omega)$ by

$$(4.14) \quad \mathcal{J}g(x, \lambda) = \left(\frac{r}{\pi}\right)^n \int e^{2r\psi(x, \bar{y})} a(x, \bar{y}, r) g(y, \lambda) e^{-2r\phi(y)} L(dy),$$

where a is a classical analytic symbol of order 0. Then, as we saw in [24] if $C_0 > 0$ is large enough and $\bar{\phi}(x) = \phi(x) - d(x, \partial\Omega)^2/C_0$ there is a constant $C > 0$ such that

$$(4.15) \quad \|g - \mathcal{J}g\|_{H_{\bar{\phi}}(\Omega)} \leq C \|g\|_{H_{\phi}(\Omega)} \quad r \gg 1$$

(for all $g \in H_{\phi}(\Omega)$). In the definition of $\mathcal{J}g$ we have the $H_{\bar{\phi}}$ -scalar product between g and an element of H_{ϕ} . Hence $\mathcal{J}g = \mathcal{J}f$ when $g = Bf$, so (4.15) gives

$$(4.16) \quad \|Bf - \mathcal{J}f\|_{H_{\bar{\phi}}(\Omega)} \leq C \|Bf\|_{H_{\phi}(\Omega)} \leq C \|f\|_{L^2_{\phi}(\Omega)}.$$

We also note that $\partial_{\bar{r}} B = B(-2\phi) - B(-2\phi)B$.

If $Gv = B(1_{\Omega \setminus \Omega_0} Dv)$ then in view of the inequality on page 161 in [24] it follows that there exists a continuous function $\tilde{\phi} < \phi$ on Ω_0 such that $G_{\lambda} = \mathcal{O}(|\lambda|)$, $\nabla G_{\lambda} = \mathcal{O}(|\lambda|^3)$ as operators $H_{\phi}(\Omega) \rightarrow H_{\tilde{\phi}}(\Omega_0)$. Here we also used that the norms of D and $\nabla_{\lambda} D$ are $\mathcal{O}(|\lambda|)$ and $\mathcal{O}(|\lambda|^3)$ resp.. This will be used with (4.13) which we write $PCv = v - Gv$.

Let now $v = \sum_{k=0}^{\infty} v_k(x) \lambda^{-k}$ be a formal analytic symbol. If $\chi \in C_0^{\infty}(\mathbb{C})$ is equal to 1 near 0, we can then choose a realisation of v (denoted by the same letter)

$$v(x, \lambda) = \sum v_k(x) \lambda^{-k} \chi(k/C\lambda) \quad |\lambda| \gg 1$$

with C sufficiently large. (cf. [24], section 1. Here λ is complex). Then $v = \mathcal{O}(1)$ and with a new constant $C > 0$ we have

$$(4.17) \quad \bar{\partial}_{\lambda} v = \mathcal{O}(e^{-C|\lambda|})$$

in the sup norm on Ω . We now choose $\phi = \frac{1}{2} |x|^2$ again and put $u = Cv$, so that

$$(4.18) \quad Pu = v - Gv \text{ in } \Omega_0, \quad u = \mathcal{O}(|\lambda|) \text{ in } H^{\phi}(\Omega),$$

which implies

$$(4.19) \quad P(\bar{\partial}_{\lambda} u) = \bar{\partial}_{\lambda} v - \frac{\partial G}{\partial \bar{\lambda}} v - G \frac{\partial v}{\partial \bar{\lambda}} - \frac{\partial P}{\partial \bar{\lambda}} u = w \text{ in } \Omega_0.$$

Since $v = \mathcal{O}(1)$ in $H_{\phi}(\Omega)$, we have

$$(4.20) \quad \frac{\partial G}{\partial \bar{\lambda}} v - G \frac{\partial v}{\partial \bar{\lambda}} = \mathcal{O}(|\lambda|^3) \text{ in } H_{\tilde{\phi}}(\Omega_0).$$

Using Stokes' formula in the form of Lemma 12.2 of [24] on the contour integral (4.3) after introduction of a suitable cut off function $\chi(y)$, it is easy to show (cf. [24], section 10, section 12) that $\frac{\partial P}{\partial \bar{\lambda}}$ is $\mathcal{O}(1)$ as operator $H_{\phi}(\Omega) \rightarrow H_{\tilde{\phi}}(\Omega_0)$, provided that $\tilde{\phi} < \phi$ is not too small. We conclude that

$$(4.21) \quad w = \mathcal{O}(|\lambda|^3) \text{ in } H_{\tilde{\phi}}(\Omega_0).$$

We now use the fact that the estimate (4.10) is valid also for P and for suitable small perturbations of ϕ such as $\phi_{\epsilon}(x) = \phi(x) + \epsilon|x|^2$. Choosing $\epsilon > 0$ sufficiently small and with $\Omega_1 \subset \subset \Omega_0$ we get

$$(4.22) \quad \|\bar{\partial}_{\lambda} u\|_{H_{\phi_{\epsilon}}(\Omega_1)} \leq C(\|w\|_{H_{\phi_{\epsilon}}(\Omega_0)} + \|\bar{\partial}_{\lambda} u\|_{H_{\phi_{\epsilon}}(\Omega_0 \setminus \Omega_1)}).$$

The right hand side is exponentially decreasing. More precisely, if $\tilde{\phi} < \phi$ is not too small, we have

$$(4.23) \quad \|\bar{\partial}_{\lambda} u\|_{H_{\phi}(\Omega_1)} = \mathcal{O}(1), \quad |\lambda| \gg 1.$$

So far we have only constructed u for $|\lambda| \geq 1$, we extend u to $|\lambda| \leq 1$ so as to stay C^∞ in λ and holomorphic in x .

Proposition 4.3. *In the above situation let us denote by v_f the formal analytic symbol corresponding to v (and which we denoted first by v), and let $u_f = \sum_0^{\infty} u_j(x) \lambda^{-j}$ be the unique formal symbol such that $P u_f = v_f$. Then u_f is a formal analytic symbol and u is a realization of u_f in a neighborhood of 0.*

Proof : Let $u_f^N = \sum_0^N u_j(x) \lambda^{-j}$. Then $P u_f^N = v + \mathcal{O}_N(\lambda^{-N})$ uniformly in Ω_0 and hence also in $H^\Phi(\Omega_0)$. The a priori estimate (4.10) with P^* replaced by P , Ω_0 by Ω_1 and Ω by Ω_0 , then shows that $u_f^N - u = \mathcal{O}(\lambda^{-N})$ in $H_\Phi(\Omega_1)$ (since $\Phi \geq 0$ with strict inequality outside 0). It follows that

$$(4.24) \quad u^{(\alpha)}(0, \lambda) = \sum_0^N u_j^{(\alpha)}(0) \lambda^{-j} + \mathcal{O}_{\alpha, N}(\lambda^{-(N+1)})$$

for all $\alpha \in \mathbb{N}^n$, $N \in \mathbb{N}$.

Lemma 4.4. *Possibly after decreasing Ω_1 around the origin we have*

$$(4.25) \quad u_N(x) = \lim_{R \rightarrow \infty} \frac{1}{2\pi i} \int_{|\lambda|=R} \lambda^{N-1} u(x, \lambda) d\lambda = \frac{1}{2\pi i} \oint_C \lambda^{N-1} \frac{\partial u}{\partial \lambda} d\lambda \wedge dx$$

for all $x \in \Omega_1$.

Proof of the Lemma : Since $\frac{\partial u}{\partial \lambda} = \mathcal{O}(e^{-C_0|\lambda|})$ uniformly for $x \in \Omega_1$, when Ω_1 is small enough by (4.23) the last integral converges for $x \in \Omega_1$, and Stokes' formula shows that the last equality holds. To show the first equality it is enough to prove the identities

$$(4.26) \quad u_N^{(\alpha)}(0) = \lim_{R \rightarrow \infty} \frac{1}{2\pi i} \int_{|\lambda|=R} \lambda^{N-1} u^{(\alpha)}(0, \lambda) d\lambda,$$

which are evident by (4.24).

Applying (4.25) and the exponential decrease of $\frac{\partial u}{\partial \lambda}$ in Ω_1 , we get

$$|u_N(x)| < C \iint |\lambda|^{N-1} e^{-C_0|\lambda|} d\lambda \wedge dx$$

$$C_1 \int_0^\infty e^{-C_0 t} t^N dt \leq C_1 C_0^{1-N} N!, \quad x \in \Omega_1,$$

which shows that u_f is a formal analytic symbol in Ω_1 . If $\tilde{u}$ is a suitable realization, then $P\tilde{u} = v$ up to an exponentially small error near the origin and applying again our a priori inequalities we see that $u - \tilde{u}$ is exponentially small when x is in a neighborhood of the origin. This proves the Proposition.

#

We have now proved the hard part of Theorem 4.1, namely the fact that if $Pu = v$ and v is a formal analytic symbol and u a formal symbol, then u is a formal analytic symbol near the origin. Outside the origin the problem is of principal type or "strictly hyperbolic" and it is well known see for instance [24], Th. 9.3, that the fact u is an analytic symbol propagates along the integral curves of the vector field v . In the half order case we write $v = v_1 + \lambda^{-1/2} v_2 \dots$. The proof is complete.

#

We now fix an eigenvalue E_0 as in section 3 and notice that $e_1, \dots, e_{N_0}$ are also analytic symbols. Since we also have a certain uniformity in Theorem 4.1, we see that Π (given by (3.23)) also conserves analytic symbols, hence $\Pi(e_j) \in \mathcal{F}_a^{m_j, k_j}(\Omega)$, with $m_j + k_j/2 = 0$. Since analytic symbols give rise to analytic symbols in the stationary phase expansions we know that $(v_1 | v_2)$ in (3.14) is an analytic symbol if v_1 and v_2 are. The matrix A introduced after (3.25) is therefore an analytic symbol and so is B since it is obtained by an elliptic construction from A . (The easiest way to see this, is to choose a double valued realization of A , which gives rise to an asymptotic sum $\tilde{B}$ of B such that $\bar{\partial}_\lambda \tilde{B}$ is exponentially decreasing, and then conclude by a modification of the argument of the end of the proof of Proposition 4.3.). The symbols $f_1, \dots, f_{N_0}$ are therefore analytic and so is the matrix Q in (3.26). We now work in the basis $f_1, \dots, f_{N_0}$ and with the matrix Q , which is double valued and hermitian for $\lambda \geq 0$. We also know that Q is an analytic symbol of order $-1/2$. Let $\lambda_1 \in S_{1/2}^{-1/2}(\mathbb{R}_+)$ be a formal eigenvalue of multiplicity d as in the proof of Theorem 3.6. Let $\tilde{\lambda}_1$ be an asymptotic sum, that we can take to be a double valued smooth function on $\mathbb{C}$. Let $\tilde{Q}$ be a realization of Q , i.e. an asymptotic sum of Q such that

$\tilde{Q}_\lambda = \mathcal{O}(e^{-|\lambda|C})$. We may choose $\tilde{Q}$ to be hermitian for $\lambda > 0$. Then if $N_1 \in \mathbb{N}$ is sufficiently large, for $|\lambda|$ sufficiently large, $\tilde{Q}$ has precisely d eigenvalues inside the disc $|E - \tilde{\lambda}_1| \leq |\lambda|^{-N_1}$ and they are all asymptotic to $\tilde{\lambda}_1$, while the other eigenvalues are at a distance $\geq |\lambda|^{-N_1+1}$ from $\tilde{\lambda}_1$. Let

$$\tilde{\Pi}_1 = \bigoplus_{|E - \tilde{\lambda}_1| \leq |\lambda|^{-N_1}} (E - \tilde{Q})^{-1} dE$$

be the spectral projection associated to these d eigenvalues. As we saw in the proof of Theorem 3.6, this double valued C^∞ function of λ defined for $|\lambda|$ sufficiently large has an asymptotic expansion Π_1 in powers of $\lambda^{-1/2}$ (which is the formal spectral projection of Q associated to λ_1) actually of order 0 because of the selfadjointness for real positive values of λ . It is also easy to see that $\tilde{\Pi}_1 = \mathcal{O}(e^{-|\lambda|C})$ and it follows by a modification of the argument in the end of the proof of Proposition 4.3, that Π_1 is an analytic symbol with $\tilde{\Pi}_1$ as a realization. Since $\lambda_1 = \frac{1}{d} \text{trace}(\Pi_1 Q)$, it follows that λ_1 is an analytic symbol and similarly for the other eigenvalues. It is also clear that we can get an O.N. basis of eigenvectors which are analytic symbols. We get

Theorem 4.5. The g_j and the corresponding eigenvalues in Theorem 3.6. can be chosen to be analytic symbols.

Actually we worked in this section with the weight $\frac{1}{2} |z|^2$ rather than $\frac{1}{4} |z|^2$ but there is an obvious change of variables for which we can pass from one to the other. Since analytic symbols are preserved by FBI-transforms and their inverses, we get

Theorem 4.6. The a_j and E_j in Theorem 3.7 can be chosen to be analytic symbols. If $\tilde{a}_j$ and $\tilde{E}_j$ are realizations and $\tilde{u}_j = \tilde{a}_j e^{-\lambda \psi}$, then there is a constant $C > 0$ such that

$$(4.27) \quad \lambda P \tilde{u}_j - \tilde{E}_j \tilde{u}_j = \mathcal{O}(e^{-\lambda/C})$$

uniformly for $\lambda \geq 1$, $|x| \leq 1/C$.

Again we may take $a_j, \tilde{a}_j, \tilde{u}_j$ real valued.

In the one-dimensional case F. Pham has recently and independently proved Theorem 4.6. (See "Calcul microdifférentiel complexe et méthode semi-classique", 30ème rencontre entre physiciens théoriciens et mathématiciens, R.C.P. n° 25, I.R.M.A. Strasbourg, Mai 1983).

5. THE INTERACTION MATRIX IN THE CASE OF STRICT MINIMA

The first part of this section is devoted to various estimates for a fixed $M_j = M_1$ under the same assumptions as in section 3, sometimes also with the analyticity assumptions. The first task will be to show that the asymptotic eigenvalues of the preceding sections are close to the true eigenvalues and similarly for the eigenfunctions in L^2 norm. After that we shall show (under a suitable additional hypothesis, that the approximate eigenfunctions are close to the true ones far away from $U_j = \{0\}$, even when we take into account the exponential decrease factor.

Let $R_0 > 0$ be some fixed value that avoids $\text{Sp}(Q_0)$. We shall only consider eigenvalues E in the interval $I(h) = [0, R_0 h]$ (or occasionally in a larger interval of the same type). If $\Omega \subset \subset M_1$ is an open sufficiently small coordinate neighborhood of 0, we have

Proposition 5.1. For every $M \in \mathbb{N}$, there is a constant $C > 0$ such that for all $E \in I(h)$, $h > 0$ sufficiently small and $u \in C_0^\infty(\Omega)$:

$$(5.1) \quad \|u\|_{M+2} \leq C_M \left(\left\| \frac{1}{h} (P-E)u \right\|_M + \|u\|_0 \right).$$

Here

$$(5.2) \quad \|u\|_M = \sum_{|\alpha| \leq M} h^{-(|\alpha| + |\beta|)/2} \|x^\alpha (hD)^\beta u\|_{L^2(\Omega)}.$$

Proof : We write

$$P = -h^2 \Delta + V = \sum_{|\alpha|=2} a_\alpha(x) (hD)^\alpha + h \sum_{|\alpha|=1} a_\alpha(x) (hD)^\alpha + V(x),$$

and put

$$P_0 = \sum_{|\alpha|=2} a_\alpha(0) (hD)^\alpha + \frac{1}{2} \langle V'(0)x, x \rangle$$

so that

$$P - P_0 = \mathcal{O}(x) (hD)^2 + \mathcal{O}(1) h(hD) + \mathcal{O}(x^3)$$

P_0 is an harmonic oscillator (unitarily equivalent to hQ_0) and it is well known that (5.1) holds for P_0 when $M=0$. For any given $\varepsilon > 0$, when Ω is small enough, we have $\frac{1}{h} \|(P-P_0)u\| \leq \varepsilon \|u\|_2$, so we get (5.1) also for P when $M=0$. We now fix Ω and suppose by induction that (5.1) holds for $0 \leq M \leq M$. We notice that

$$\|u\|_{M+3} \sim \|u\|_{M+2} + \sum_{|\alpha|=1} h^{-1/2} \|x^\alpha u\|_{M+2} + \sum_{|\beta|=1} h^{-1/2} \|hD^\beta u\|_{M+2}$$

Here we use the induction hypothesis to estimate $\|u\|_{M+2}$. Also by the induction :

$$(5.3) \quad h^{-1/2} \|x^\alpha u\|_{M+2} \leq C(h^{-1/2} \left\| \frac{1}{h} (P-E)u \right\|_0 + h^{-1/2} \|x^\alpha u\|_0)$$

the last term to the right can be estimated by

$$\|u\|_2 \leq C \left(\left\| \frac{1}{h} (P-E)u \right\|_0 + \|u\|_0 \right)$$

The first term to the right in (5.3.) can be estimated by

$$(5.4) \quad h^{-1/2} \|x^\alpha \frac{1}{h} (P-E)u\|_M + h^{-3/2} \|[P, x^\alpha]u\|_M$$

Here the first term can be estimated by $C \left\| \frac{1}{h} (P-E)u \right\|_{M+1}$. Moreover

$$[P, x^\alpha] = \mathcal{O}(1) h(hD) + \mathcal{O}(1)h^2$$

so the second term of (5.4) can be estimated by

$$C(h^{-1/2} \|hDu\|_M + h^{1/2} \|u\|_M) \leq C^2 \|u\|_{M+2} \leq$$

$$C^3 \left(\left\| \frac{1}{h} (P-E)u \right\|_M + \|u\| \right)$$

The estimate of $h^{-1/2} \|hD^\beta u\|$ is similar and we can conclude the induction step.

#

Let $u \in \mathcal{D}_{P_{M_1}}$ be a normalized eigenfunction of P_{M_1} and let $E \in I(h)$ be the corresponding eigenvalue. Let Ω be as above and choose $\chi \in C_0^\infty(\Omega)$ equal to 1 near 0. The decay results of section 2 show that the L^2 norms over a compact region outside 0 in Ω of u and ∇u are $\mathcal{O}(e^{-C\lambda})$ for some geometric constant $C > 0$. Combining this with classical a priori estimates for the Laplacian, we get the same control over $D^\alpha u$ for all $\alpha \in \mathbb{N}^n$. If $v = \chi u$, then $\|u - v\|_{L^2} = \mathcal{O}(\exp - \lambda C)$ so in particular $\|v\| \sim 1$. Moreover

$$Pv = Ev + [P, \chi]v$$

where

$$\|D^\alpha [P, \chi]v\| \leq C_\alpha e^{-\lambda C}, \quad \forall \alpha \in \mathbb{N}^n$$

Proposition 5.1. then shows that $\|v\|_{M+2} \leq C_M$ for every $M \in \mathbb{N}$. On the other hand

$$\|(P - P_0)v\| \leq \text{Const. } h^{3/2} \|v\|_3$$

so

$$(5.5) \quad P_0 v = Ev + w, \quad w = \mathcal{O}(h^{3/2}) \text{ dans } L^2$$

We conclude that there is a constant $C_0 > 0$ such that when h is sufficiently small :

$$(5.6) \quad \text{Sp}(P_{M_1}) \cap I(h) \subset \text{Sp } P_0 + B(0, C_0 h^{3/2})$$

(Again we recall that $\text{Sp } P_0 = h \text{Sp } Q_0$ is known explicitly). Let E_0 be an eigenvalue of Q_0 of multiplicity N_0 . (5.6) does not give the number of eigenvalues of P_{M_1} in the interval $E_0 h + B(0, C_0 h^{3/2})$, but without chan-

ging this number, we may deform P and the metric continuously in such a way that nothing is changed outside Ω , and such that P becomes P_0 near 0 and the metric becomes constant. Then we are essentially in a situation, where the arguments of section 2 show that the eigenvalues of P_{M_1} and P_0 agree modulo an exponentially small perturbation of the identity map. Hence the number of eigenvalues of P_{M_1} in $E_0 h + B(0, C_0 h^{3/2})$ is indeed N_0 and we get

Proposition 5.2. *There is a bijection $b: Sp(P_{M_1}) \cap I(h) \rightarrow Sp(P_0) \cap I(h)$ such that $|b(E) - E| \leq C_0 h^{3/2}$. (b is defined for h sufficiently small).*

Let $\mathcal{E}_1 \subset L^2(M_1)$ be the eigenspace of dimension N_0 associated to the N_0 eigenvalues of P_{M_1} in the interval $I_{E_0}(h) = E_0 h + B(0, C_0 h^{3/2})$. It is now clear using Proposition 2.5., that we have.

Proposition 5.3. *Let $\tilde{a}_j, \tilde{E}_j$ be given in Remark 3.10 and put $\hat{u}_j = h^{-n/4} \tilde{a}_j e^{-\psi/h} \chi$, $j = 1, \dots, N_0$. If $\mathcal{E}_1$ is the space spanned by the $\hat{u}_j$, then $\vec{d}(\mathcal{E}_1, \mathcal{E}_1) = \mathcal{O}(h^\infty)$. The eigenvalues of P_{M_1} in $I_{E_0}(h)$ are $\hat{E}_j + \mathcal{O}(h^\infty)$.*

Proposition 5.4. *In the analytic case, let $\tilde{a}_j, \tilde{E}_j$ be as in Theorem 4.6, and let $\mathcal{E}_1$ be the space spanned by $\hat{u}_j = h^{-n/4} \tilde{a}_j e^{-\psi/h} \chi$. Then there is a constant $C > 0$ such that $\vec{d}(\mathcal{E}_1, \mathcal{E}_1) = \mathcal{O}(e^{-\lambda/C})$ and such that the eigenvalues of P_{M_1} in $I_{E_0}(h)$ are $\hat{E}_j + \mathcal{O}(e^{-\lambda/C})$.*

We next start the study of the eigenfunctions far away from the well. In the C^∞ case we need to improve the decay estimates in section 2. Let $d(x, y)$ denote the distance in M for the Agmon metric and write $d(x) = d(x, 0)$.

Proposition 5.5. *There exist constants $C, \tilde{C} > 0$, such that for every normalized eigenfunction u of P_{M_1} with eigenvalue $E \in I(h)$:*

$$(5.7) \quad h^2 \left\| \left(1 + \frac{d}{h}\right)^{-C} e^{d/h} \nabla u \right\|^2 + h \left\| \left(1 + \frac{d}{h}\right)^{-C} e^{d/h} u \right\|^2 \leq \tilde{C} h.$$

Proof: We choose

$$\phi(x) = \begin{cases} d(x) - C h \log d(x)/h & \text{for } d(x) \geq C h \\ d(x) - C h \log C & \text{for } d(x) \leq C h \end{cases}$$

Then for $d(x) \leq C h$ we are in the region where $d(x)$ is smooth so $V(x) - (\nabla \phi(x))^2 = 0$, while for $d(x) \geq C h$ we have

$$\nabla \phi(x) = \left(1 - \frac{C h}{d(x)}\right) \nabla d(x) \quad \text{a.e.}$$

Hence by simple estimates,

$$V(x) - (\nabla \phi(x))^2 \geq \frac{V(x)}{d(x)} C h$$

for $d(x) \geq C h$. Now we use that $C_0^{-1} \leq V(x)/d(x) \leq C_0$ for some constant $C_0 > 0$. Hence

$$V - (\nabla \phi(x))^2 \geq \frac{C}{C_0} h \quad \text{a.e. for } d(x) \geq C h.$$

Now apply (1.3) of Theorem 1.1 for the operator $P - E$. We can take

$$F_+(x) = 1_{\{d(x) \geq C h\}} \sqrt{V - E - (\nabla \phi)^2} \geq 1_{\{d(x) \geq C h\}} \left(\frac{C}{C_0} - \frac{E}{h}\right)^{1/2} h^{1/2}$$

$$F_-(x) = 1_{\{d(x) \leq C h\}} E^{1/2}$$

and we require that $\frac{C}{C_0} \geq \frac{E}{h} + 1$. Then (1.3) gives

$$h^2 \|\nabla(e^{\phi/h} u)\|^2 + \frac{1}{2} \left(\frac{C}{C_0} h - E\right) \|e^{\phi/h} u\|_{\{d \geq C h\}}^2 \leq \frac{3}{2} E \|e^{\phi/h} u\|_{\{d \leq C h\}}^2$$

which we write as

$$(5.8) \quad h^2 \|\nabla(e^{\phi/h} u)\|^2 + \frac{1}{2} \left(\frac{C}{C_0} h - E\right) \|e^{\phi/h} u\|^2 \leq \left(\frac{h}{2} \frac{C}{C_0} + E\right) \|e^{\phi/h} u\|_{\{d \leq C h\}}^2$$

To conclude it suffices to notice that $e^{\phi/h}$ is of the same order of magnitude as $(1 + d(x)/h)^{-C} e^{d(x)/h}$, and that

$$h \|e^{\Phi/h} \nabla u\| \leq h \| \nabla (e^{\Phi/h} u) \| + \text{const. } h^{1/2} \| (\frac{d}{h})^{1/2} e^{\Phi/h} u \|$$

#

Let $u_1, \dots, u_{N_0} \in \mathcal{E}_1$ be an O.N. basis of eigenfunctions and let the corresponding eigenvalues be $E_1, \dots, E_{N_0}$. From Proposition 5.3 and 5.4 it follows that there is an orthogonal matrix $(c_{j,k}^{(h)})_{1 \leq j,k \leq N_0}$ such that in $L^2(M_1)$:

$$(5.9) \quad u_j = \sum_k c_{j,k} \hat{u}_k + \mathcal{O}(h^\infty)$$

$$(5.10) \quad \hat{u}_j = \sum_k c_{j,k} \hat{u}_k + \mathcal{O}(e^{-\lambda/C}) \quad (\text{analytic case})$$

Moreover we may choose $c_{j,k}$ such that $c_{j,k} = 0$ when $\hat{E}_k(\tilde{E}_k)$ is not asymptotically equal to E_j . Naturally in the case when the $\hat{E}_j(\tilde{E}_j)$ all have different asymptotic expansions, then we may choose $(c_{j,k})$ to be the identity matrix.

Let $\Omega \subset \subset M_1$ be open containing 0 and assume

$$(5.11) \quad d(x) \text{ is a } C^2 \text{ function on } \Omega.$$

Then $\Lambda_+ = \{(x, \nabla d(x)) ; x \in \Omega\} \subset q^{-1}(0)$, where $q(x, \xi) = \xi^2 - V(x)$ so H_q is tangent to Λ_+ . We also assume

$$(5.12) \quad \text{If } (x, \xi) \in \Lambda_+ \text{ then } \exp t H_q(x, \xi) \in \Lambda_+ \text{ for } t \leq 0 \text{ and } \exp t H_q(x, \xi) \rightarrow (0, 0) \text{ when } t \rightarrow -\infty.$$

Since we already know that Λ_+ is C^∞ (resp. analytic) near $(0, 0)$ we have the same regularity everywhere and hence

$$(5.13) \quad d(x) \text{ is a } C^\infty \text{ (resp. analytic) function on } \Omega.$$

We notice as in the beginning of section 3 that $d(x)$ is the length of the curve $]-\infty, 0[\ni t \rightarrow \Pi_x \exp t H_q(x, \nabla d(x)) = x_t(x)$ for the metric $V dx^2$.

From now on we assume that all the wells are points $U_j = \{x_j\}$ and that

$$(5.14) \quad V''(x_j) > 0, \quad 1 \leq j \leq N.$$

Lemma 5.6. Let $x, y \in \Omega$ and assume that y is not on the segment $\{0\} \cup \{x_t(x) ; -\infty < t \leq x\}$. Then $d(0, x) < d(0, y) + d(y, x)$.

Proof : Let $\tilde{M}$ be M equipped with the Agmon metric $V dx^2$. By an $\tilde{M}$ -geodesic we mean a (continuous) curve $[a, b] \ni t \rightarrow x(t) \in \tilde{M}$, which up to reparametrization is a geodesic outside the U_j . Such a geodesic has a well defined length in $[0, +\infty)$ and if $x, y \in \tilde{M}$, then there exists an $\tilde{M}$ -geodesic, joining x and y , of length $d(x, y)$. We call it a minimal geodesic. Now let $x, y \in \Omega$ be as in the lemma. Let $\gamma_0 : [0, 1] \ni t \rightarrow \Omega$ be the segment $\{0\} \cup \{x_t(x) ; -\infty < t \leq 0\}$ reparametrized so that $\gamma_0(0) = x$, $\gamma_0(1) = 0$. γ_0 is then a minimal $\tilde{M}$ -geodesic and we claim that it is the unique (up to reparametrization) minimal $\tilde{M}$ -geodesic joining 0 and x . Indeed, suppose that $\gamma : [0, 1] \rightarrow \tilde{M}$ is another one. Then since d is of class C^1 near x and $d(x) \neq 0$ we know that $\langle \nabla d(x), \dot{\gamma}(0) \rangle = - \|\dot{\gamma}(0)\| \|\nabla d\|, \langle \nabla d(x), \dot{\gamma}_0(0) \rangle = - \|\dot{\gamma}_0(0)\| \|\nabla d\|$ and we conclude that $\dot{\gamma}(0) = k \dot{\gamma}_0(0)$, $k > 0$. Hence γ coincides with γ_0 all the way to 0.

By the triangle inequality $d(0, x) \leq d(0, y) + d(y, x)$. If we have equality then a minimal $\tilde{M}$ -geodesic from x to y compared with the minimal $\tilde{M}$ -geodesic from y to 0 would be a minimal $\tilde{M}$ -geodesic from x to 0 distinct from γ_0 , hence a contradiction, and the lemma follows.

#

From now on we use the term "geodesic" instead of " $\tilde{M}$ -geodesic". An easy compactness argument gives :

Lemma 5.7. Let $K_1, K_2 \subset \Omega$ be compact and such that K_2 is disjoint from the compact union $\hat{K}_1$ of all minimal geodesics from all points of K_1 to 0. Then there exists $\varepsilon > 0$ such that for all $x \in K_1, y \in K_2$ we have

$$(5.15) \quad d(0, x) < (1 - \varepsilon) (d(0, y) + d(y, x)).$$

Theorem 5.8. Let $\Omega \subset \subset M_1$ be an open set satisfying (5.11), (5.12). Then if we put $\psi = d(x)$, the various symbols $a, \hat{a}, \hat{a}$ of Remark 3.10, Theo-

rem 4.6, Proposition 5.3 and 5.4 are naturally defined in Ω , by solving the transport equations globally, and hence also the functions $\hat{u}_k$ in (5.9), (5.10). If u_j and c_{jk} are as in those equations and $u_j' = \sum c_{jk} \hat{u}_k$ then :

(A) For every compact $K \subset \Omega$ and every $N \in \mathbb{N}$ we have

$$(5.16) \quad \| \nabla(e^{d/h}(u_j - u_j')) \|_{L^2(K)} + \| e^{d/h}(u_j - u_j') \|_{L^2(K)} = \mathcal{O}(h^N).$$

(B) In the analytic case we can replace the right hand side of (5.15) by $\mathcal{O}(e^{-\lambda/C})$ where $C > 0$ may depend on K .

Proof : The first remarks are obvious from the geometric assumptions and we shall first give the proof of (A) and then the proof of (B) which is somewhat simpler. Let $w = u_j - u_j'$. Then from Proposition 5.5 and the more explicit form for u_j' , we see that the left hand side of (5.16) is of at most polynomial growth in $\lambda = 1/h$. Also

$$(5.17) \quad Pw - E_j w = r,$$

where $r = (P - E_j)u_j$ has the property that

$$(5.18) \quad \| e^{d/h} r \|_{L^2(K)} = \mathcal{O}(h^N),$$

for every $N \in \mathbb{N}$ and every compact $K \subset \Omega$. (5.9) gives

$$(5.19) \quad \| w \|_{L^2(K)} = \mathcal{O}(h^N).$$

Now fix a compact set $K \subset \Omega$ and let $\chi \in C_0^\infty(\Omega)$ be equal to 1 in a neighborhood of the union $\hat{K}$ of all minimal geodesics from points of K to 0. If ϕ is the weight function, used in the proof of Proposition 5.5, we shall here apply Theorem 1.1 with $u = \chi w$, with P replaced by $P - E_j$ and with the new weight function

$$(5.20) \quad \phi_N(x) = \min(\phi(x) + Nh \log 1/h, \inf_{y \in \text{supp } \nabla \chi} (\phi(y) + (1-\epsilon) d(x, y))).$$

Here $\epsilon > 0$ will be small enough, depending on χ and K , but not on N which will be chosen arbitrarily large.

By Lemma 5.7, if W is a small neighborhood of $\hat{K}$ and $\epsilon > 0$ sufficiently small, then for every $N \in \mathbb{N}$ there exists $h_N > 0$ such that $h < h_N$, $x \in W \Rightarrow \phi_N(x) = \phi(x) + Nh \log 1/h$. On the other hand, if $\psi(x) = \inf_{\text{supp } \nabla \chi} \phi(y) + (1-\epsilon)d(x, y)$, then $V(x) - (\nabla \psi)^2 \geq (2\epsilon - \epsilon^2)V(x) \geq (\text{const.})_\epsilon$ in a region bounded away from 0. If C is the same constant as in the proof of Proposition 5.5, we conclude that (a.e.) :

$$(5.21) \quad V(x) - (\nabla \phi_N(x))^2 \begin{cases} \geq Ch/C_0 & \text{for } d(x) \geq Ch \\ = 0 & \text{for } d(x) \leq Ch \end{cases}$$

We choose again $\frac{C}{C_0} \geq \frac{E_j}{h} + 1$ and

$$F_+ = 1_{\{d(x) \geq Ch\}} \sqrt{W - E_j - (\nabla \phi_N)^2} \geq 1_{\{d \geq Ch\}} \left(\frac{C}{C_0} h - E_j \right)^{1/2}$$

$$F_- = 1_{\{d \leq Ch\}} E_j^{1/2}.$$

Then

$$(5.22) \quad h^{2-2N} \| \nabla(e^{\phi/h} w) \|_{L^2(K)}^2 + h^{1-2N} \| e^{\phi/h} w \|_{L^2(K)}^2 \leq C_1 (h^{1-2N} \| e^{\phi/h} w \|_{L^2(d \leq Ch)}^2 + h^{-1} \| e^{\phi_N/h} (P - E_j) \chi w \|^2).$$

Here the first term to the right is $\mathcal{O}(1)$, $h \rightarrow 0$ by (5.19). The last term to the right can be estimated by

$$(5.23) \quad 2h^{-1} \| e^{\phi_N/h} \chi (P - E_j) w \|^2 + 2h^{-1} \| e^{\phi_N/h} [P, \chi] w \|^2 \leq 2h^{-(1+2N)} \| e^{\phi/h} \chi (P - E_j) w \|^2 + 2h^{-1} \| e^{\phi/h} [P, \chi] w \|^2,$$

since $\phi_N \leq \phi$ on $\text{supp } \nabla \chi$. Here the first term to the right is $\mathcal{O}(1)$ by

(5.17), (5.18), and the second term is $\mathcal{O}(h^{-N_0})$ for some fixed N_0 . Applying these bounds to the right in (5.22) we get (A).

In the analytic case the proof is simpler. In (5.18), (5.19) we can replace $\mathcal{O}(h^N)$ by $\mathcal{O}(e^{-\lambda C_K})$. We replace Φ_N by

$$(5.24) \quad \tilde{\Phi}(x) = (1 - \epsilon) \min(d(x) + a, \inf_{y \in \text{Supp } \nabla x} d(y) + d(y, x))$$

Then, if $a > 0$ is small enough we have $\tilde{\Phi}(x) = (1 - \epsilon)(d(x) + a)$ in W and now $V(x) - (\nabla \tilde{\Phi})^2 \geq (\text{const.})\epsilon$ in any region, bounded away from 0. Thus for every $\epsilon > 0$ and $r > 0$ we can choose $F_+ \geq \text{const.} > 0$ (everywhere), $\text{supp } F_- \subseteq \{x : d(x) \leq r\}$. Then (1.3) implies

$$(5.25) \quad (h^2 \|\nabla(e^{(1-\epsilon)d/h} w)\|_{L^2(K)}^2 + \|e^{(1-\epsilon)d/h} w\|_{L^2(K)}^2) e^{2(1-\epsilon)a/h} \\ \leq C_{r, \epsilon, a} (\|e^{(1-\epsilon)(d+a)/h} w\|_{L^2(d(x) \leq r)}^2 + \|e^{\tilde{\Phi}/h} \chi_{(P-E_j)W}\|^2 \\ + \|e^{\tilde{\Phi}/h} [P, \chi] w\|^2).$$

If we choose a and r sufficiently small, then by (5.10) the first term to the right in $\mathcal{O}(1)$ as $h \rightarrow 0$ (for all $\epsilon > 0$). We estimate the second term to the right by

$$e^{2(1-\epsilon)a/h} \|e^{d/h} \chi_{(P-E_j)W}\|_{L^2}^2 = \mathcal{O}(1), \quad h \rightarrow 0$$

if $a > 0$ is sufficiently small. Finally the third term can be estimated by some constant since $\tilde{\Phi} < d$ in $\text{supp } \nabla x$. The right hand side of (5.24) is therefore $\mathcal{O}_\epsilon(1)$ if we fix $a > 0$ (and $r > 0$) sufficiently small but independent of ϵ . Then we get (B) by taking $\epsilon > 0$ sufficiently small.

#

As before we let S_0 denote $\min_{j \neq k} d(x_j, x_k)$. Let $\Omega_j \subset M$ be open neighborhoods of x_j with $x_k \notin \Omega_j$ when $k \neq j$ and assume that $\Omega = \Omega_j$ satisfies (5.11), (5.12) for each j (with $d(x) = d(x, x_j) = \psi_j(x)$). Notice that if $x, y \in M$ then there

is a geodesic (for $\forall dx^2$) joining x and y and whose length is $d(x, y)$. Such a geodesic is automatically C^∞ except possibly at the points $x_1, \dots, x_N$. (Here we do not care about the choice of parametrization). Now assume

$$(5.26) \quad \text{If } d(x_j, x_k) = S_0, \text{ then there is a unique geodesic } \gamma_{j,k} \text{ of length } d(x_j, x_k), \text{ joining } x_j \text{ and } x_k.$$

$$(5.27) \quad \text{With } \gamma_{j,k} \text{ as above we have } \gamma_{j,k} \subset \Omega_j \cup \Omega_k.$$

$$(5.28) \quad \text{Let } x_0 \in \gamma_{j,k} \cap \Omega_j \cap \Omega_k \text{ and let } \Gamma \subset \Omega_j \cap \Omega_k \text{ be a smooth hypersurface which intersects } \gamma_{j,k} \text{ transversally at } x_0 \text{ and such that this is the only point of } \Gamma \cap \gamma_{j,k}. \text{ Then there is a constant } C > 0 \text{ such that}$$

$$d(x, x_j) + d(x, x_k) \geq S_0 + \frac{1}{C} d(x, x_0)^2$$

for all $x \in \Gamma$.

It is an easy exercise to show that the choice of x_0 and Γ does not affect the validity of (5.28). We choose M_j as before with $S > S_0/2$, $\tilde{M}_j \supset \Omega_j$.

Let E_0 be in the spectrum of some of the localized harmonic oscillators and let $E(h) = E_0 h + \mathcal{O}(h^{3/2})$ be one of the asymptotic eigenvalues that we have constructed above for the various wells. If we choose $I(h) = [E(h) - h^{N_0}, E(h) + h^{N_0}]$, with N_0 sufficiently large then (with the notation of section 2) the only μ_α that appear in $I(h)$ are equal to $E(h) + \mathcal{O}(h^N)$ for all N . It will be convenient to assume

$$(5.29) \quad \text{If } j(\alpha) = j(\beta) \text{ then } \mu_\alpha = \mu_\beta.$$

Then the relevant parts of the matrices $c_{j,k}$ in (5.9), (5.10) may be chosen to be the identity. Naturally (5.29) is much stronger than equality between the asymptotic eigenvalues from the same well, but if our operator is invariant under certain groups of isometries, then with a convenient choice of $M_j(\alpha)$, the condition seems to be realistic.

In the analytic case we notice that there is a constant $b > 0$ such that

$$(5.30) \quad |\mu_\alpha - \mu_\beta| = \mathcal{O}(e^{-b/h}), \quad \forall \alpha, \beta.$$

Let Γ be as in (5.28). Then by Remark 2.13, we have when

$$d(x_{j(\alpha)}, x_{j(\beta)}) = S_0;$$

$$(5.31) \quad \hat{w}_{\alpha, \beta} \equiv h^2 \int_{\Gamma} \varphi_\alpha \frac{\partial \varphi_\beta}{\partial n} - \varphi_\beta \frac{\partial \varphi_\alpha}{\partial n} dS,$$

where n is the unit normal of Γ oriented "towards" $x_{j(\alpha)}$ in the obvious sense. Here " $\equiv$ " denotes equality modulo a term which is $\mathcal{O}(h^N e^{-S_0/h})$ for all $N \in \mathbb{N}$ in the C^∞ case and modulo a term which is $\mathcal{O}(e^{-(S_0+a)/h})$ for some $a > 0$ in the analytic case.

We next apply Theorem 5.8 (recalling that we can take $c_{jk} = \delta_{j,k}$ in (5.9), (5.10)). If we denote the amplitudes appearing in Proposition 5.3 and 5.4 simply by a_α , we get by substitution in (5.31)

$$(5.32) \quad \hat{w}_{\alpha, \beta} \equiv h^{1-N/2} \int_{\Gamma} a_{\alpha, \beta} e^{-(\psi_j(\alpha) + \psi_j(\beta))/h} dS,$$

where

$$(5.33) \quad a_{\alpha, \beta} = a_\alpha a_\beta \left(\frac{\partial \psi_\alpha}{\partial n} - \frac{\partial \psi_\beta}{\partial n} \right) + h \left(a_\alpha \frac{\partial a_\beta}{\partial n} - a_\beta \frac{\partial a_\alpha}{\partial n} \right).$$

In view of (5.28) $\hat{w}_{\alpha, \beta}$ has a developpment by the method of stationary phase (see [24], section 2 for the analytic case) and we find

$$(5.34) \quad \hat{w}_{\alpha, \beta} \equiv b_{\alpha, \beta}(h) e^{-S_0/h}, \quad b_{\alpha, \beta} = b_{\beta, \alpha},$$

where $b_{\alpha, \beta}$ is a symbol (analytic symbol) whose order in $\lambda = 1/h$ is bounded by $d^0 a_\alpha + d^0 a_\beta + 1/2$. Here the orders $d^0 a_\alpha$, $d^0 a_\beta$ can be computed in principle. Adding a few more simple observations, we can resume our computations in the

Theorem 5.9. Choose $I(h)$ as above and assume (5.14), (5.26)–(5.29). Let $\hat{w}_{\alpha, \beta}$ be given by Theorem 2.10. Then for $d(x_{j(\alpha)}, x_{j(\beta)}) = S_0$ these coefficients are given by (5.34) up to an error which in the C^∞ case is $\mathcal{O}(h^N e^{-S_0/h})$ for every $N \in \mathbb{N}$ and in the analytic case is $\mathcal{O}(e^{-(S_0+a)/h})$ for some $a > 0$.

Here in the two cases $b_{\alpha, \beta}$ is a classical respectively classical analytic symbol of the form

$$(5.35) \quad b_{\alpha, \beta}(\lambda) \sim \lambda^{m(\alpha, \beta)} \sum_{j \in \frac{1}{2}\mathbb{N}} b_{\alpha, \beta}^{(j)} \lambda^{-j}$$

where the order $m(\alpha, \beta)$ can be estimated explicitly. In the special case when E_0 can appear in the spectrum of the various localized harmonic oscillators, only as the principal eigenvalue then the $b_{\alpha, \beta}$ are all elliptic symbols, strictly negative and of degree $m(\alpha, \beta) = -1/2$. In that case the sum in (5.35) is over $\mathbb{N}$.

Clearly the various leading symbols could be computed by solving transport equations. In this way one can check directly that the leading term in $b_{\alpha, \beta}$ does not depend on the choice of Γ .

Remark 5.10. When P is invariant under some group of isometries, the interaction matrix will have a corresponding invariance property and the splitting of the eigenvalues can be studied quite explicitly in many situations, as we shall see in a future paper.

Remark 5.11. In the analytic case we have stronger remainder estimates and it should be possible to study small perturbations of symmetric situations. For instance it seems possible to include in Theorem 5.9 (after some modification of the hypotheses) the description of $\hat{w}_{\alpha, \beta}$ when $d(x_{j(\alpha)}, x_{j(\beta)})$ is larger than, but very close to S_0 .

Remark 5.12. In the one-dimensional case it is not necessary to stay "near the bottom" to get explicit results. It is enough to assume that $V(x) = 0 \Rightarrow V'(x) \neq 0$. Then in principle the φ_α can be computed by the WKB or Maslov methods. We have chosen to develop first a multidimensional case since the geometry becomes much more interesting.

6. THE CASE WHEN (5.26)–(5.28) ARE NOT NECESSARILY SATISFIED

In the first part of this section we recall a result on the non-degenerate behaviour near the interior of a minimal geodesic. At least in a slightly different form this result is well known, as it has been confir-

med to us by Prof. Kobayashi. However for non specialists (such as the authors) it might be useful to give a proof based only on a short introductory text such as Milnor [19].

Let M be a complete Riemannian manifold and $N \subset M$ a compact submanifold. We consider a geodesic $\gamma: [0,1] \rightarrow M$ with $\gamma(0) \in N$, $\dot{\gamma}(0) \in TN^\perp$. Let $\Gamma: [0,1] \times [-\epsilon, \epsilon]^2 \rightarrow M$ be a variation of γ . By this we mean a continuous map such that $\Gamma(t, 0, 0) = \gamma(t)$, such that $\frac{\partial^{k+l}\Gamma}{\partial u^k \partial v^l}$ exist and are continuous and such that Γ is C^∞ on $[t_j, t_{j+1}] \times]-\epsilon, \epsilon[^2$, where $0 = t_0 < t_1 < \dots < t_N = 1$ is a finite subdivision of $[0,1]$. Let

$$(6.1) \quad E(\Gamma) = E(\Gamma)(u, v) = \frac{1}{2} \int_0^1 \left\| \frac{\partial \Gamma}{\partial t}(t, u, v) \right\|^2 dt.$$

Then using various identities for covariant derivatives and the curvature tensor (see Milnor [19]) one gets

Proposition 6.1. Let γ be as above and Γ a variation with $\Gamma|_{t=0} \in N$, $\Gamma|_{t=1} = \gamma(1)$. Then for $u = v = 0$ we have

$$(6.2) \quad \frac{\partial E}{\partial u} = 0$$

$$(6.3) \quad \frac{\partial^2 E}{\partial u \partial v} = \int_0^1 \left\langle \frac{D}{dt} \frac{\partial \Gamma}{\partial u}, \frac{D}{dt} \frac{\partial \Gamma}{\partial v} \right\rangle dt + \int_0^1 \left\langle R\left(\frac{\partial \Gamma}{\partial u}, \dot{\gamma}\right) \dot{\gamma}, \frac{\partial \Gamma}{\partial v} \right\rangle dt - \left\langle \dot{\gamma}(0), \frac{D}{dt} \frac{\partial \Gamma}{\partial v} \right\rangle \Big|_{t=0}.$$

Here $\frac{D}{dt}$ denotes covariant differentiation along γ and R the curvature tensor. Each term to the right is a symmetric bilinear form of the piecewise smooth vector fields along γ : $U = \frac{\partial \Gamma}{\partial u}$, $V = \frac{\partial \Gamma}{\partial v}$. In particular the last term is a symmetric bilinear function of $U(0)$, $V(0)$. (Here piecewise smooth means continuous and piecewise C^∞).

Naturally we have a completely analogous result, if we replace $[0,1]$ by $[0, \theta]$, $0 < \theta < 1$. We write $E_{[0, \theta]}$ for the corresponding energy function and

$$(6.4) \quad Q_\theta(U, V) = \int_0^\theta \left\langle \frac{D}{dt} U, \frac{D}{dt} V \right\rangle dt + \int_0^\theta \left\langle R(U, \dot{\gamma}) \dot{\gamma}, V \right\rangle dt + A(U(0), V(0))$$

for the corresponding hessian as in (6.3).

From now on we only consider variations with

$$(A) \quad \left(\Gamma, \frac{\partial \Gamma}{\partial t} \right) \Big|_{t=0} \in TN^\perp$$

$$(B) \quad \Gamma|_{t=0} = \gamma(0).$$

Near $\gamma(0)$ we choose local coordinates centered at $\gamma(0)$, $x = (x', x'')$ so that N is given (locally) by $x'' = 0$ and such that $TN^\perp$ is given by the $\frac{\partial}{\partial x''}$ directions. Then (A), (B)₀ imply

$$(A)' \quad \pi_{x''} U(0) = 0, \quad \pi_{x'} \dot{U}(0) = 0$$

$$(B)'_0 \quad U(0) = 0$$

and similarly for V . Here $\pi_{x''}$, $\pi_{x'}$ are the orthogonal projections onto the coordinate planes of the x'' and x' variables respectively. Conversely if U, V are piecewise smooth vector fields along γ satisfying (A)', (B)'₀, then we can find a corresponding variation satisfying (A), (B)₀.

By integration by parts one shows

Proposition 6.2. Let U, V be piecewise smooth along γ and satisfy (A)', (B)'₀. Then in the distribution sense

$$(6.4) \quad Q_\theta(U, V) = \int_0^\theta \left\langle -\frac{D}{dt} \left(\frac{D}{dt} U \right) + R(U, \dot{\gamma}) \dot{\gamma}, V \right\rangle dt.$$

We recall that the solutions of

$$(6.5) \quad -\frac{D}{dt} \left(\frac{D}{dt} U \right) + R(U, \dot{\gamma}) \dot{\gamma} = 0$$

are called Jacobi fields. They correspond to infinitesimal variations of geodesics. In our situation, if U is a Jacobi field on $[0,1]$ with $\Pi_{x_0} U(0) = 0$, $\Pi_{x_0} \dot{U}(0) = 0$, then we can find a smooth map $\Gamma : [0,1] \times]-\epsilon, \epsilon[\rightarrow M$ satisfying (A) such that $\Gamma(\cdot, u)$ is a geodesic for all u , and such that $\frac{\partial \Gamma}{\partial u} = U(0)$, $\frac{\partial^2 \Gamma}{\partial u^2} = \dot{U}(0)$ for $t = u = 0$. Then $\frac{\partial \Gamma}{\partial t} = U(t)$ for $u = 0$ and if we fix $t_0 \in [0,1]$, then the mapping $(U'(0), \dot{U}'(0)) \rightarrow U(t_0)$ can be identified with the differential at $(Y(0), \dot{Y}(0))$ of the geodesic flow map $\mathcal{H}_{t_0} : TN^{\perp} \ni (x, v) \rightarrow \exp_{t_0} v(x) \in M$. To know the differential of $\mathcal{H}_{t_0}$ at $(Y(0), \dot{Y}(0))$ is of course equivalent to knowing the differential of $\mathcal{H}_1$ at $(Y(0), t_0 \dot{Y}(0))$. We now assume the minimality condition

(6.6) The length of $Y : [0,1] \rightarrow M$ is equal to the distance from $Y(1)$ to N .

Then γ is a minimum for the energy function restricted to geodesics parametrized on $[0,1]$ from N to $Y(1)$.

Proposition 6.3. We assume (6.6). Then the differential of $\mathcal{H}_{t_0}$ at $(Y(0), \dot{Y}(0))$ is non-degenerate for every $t_0 \in]0,1[$.

Proof : Suppose the statement is wrong for some $t_0 \in]0,1[$. Then there is a Jacobi field U not identically 0 satisfying (A)' and (B)' $_{t_0}$, and hence U is in the kernel of Q_{t_0} considered as a bilinear form on $\mathcal{D}_{t_0}$; the space of piecewise smooth vector fields on $[0, t_0]$ satisfying (A)' and (B)' $_{t_0}$.

Clearly $U(t_0) \neq 0$, so if we define $\tilde{U} \in \mathcal{D}_1$ by $\tilde{U}(t) = U(t)$ for $0 \leq t \leq t_0$, $\tilde{U}(t) = 0$ for $t_0 \leq t \leq 1$, then $\tilde{U}$ is not a Jacobi field on $[0,1]$ and hence by Proposition 6.4., $\tilde{U}$ is not in the kernel of Q_1 . Take $W \in \mathcal{D}_1$ with $Q_1(\tilde{U}, W) \neq 0$. Since $Q_1(\tilde{U}, \tilde{U}) = Q_{t_0}(U, U) = 0$ we get

$$Q(\tilde{U} + \delta W, \tilde{U} + \delta W) = 2\delta Q(\tilde{U}, W) + \delta^2 Q(W, W)$$

which is < 0 if we choose δ small with a suitable sign.

If $\Gamma(t, u)$ is a variation satisfying (A), (B) $_1$ with $\frac{\partial \Gamma}{\partial u} \Big|_{u=0} = \tilde{U} + \delta W$, then

$\frac{\partial}{\partial u} E(\Gamma) = 0$, $\frac{\partial^2}{\partial u^2} E(\Gamma) < 0$ for $u = 0$ and this contradicts the minimality assumption.

≠

Under the assumptions of the preceding proposition we can find an open neighborhood $U \subset TN^{\perp}$ of $\{(Y(0), \theta \dot{Y}(0)) ; 0 < \theta < 1\}$ such that

$$(6.7) \quad (y, v) \in U, \quad 0 < \theta < 1 \Rightarrow (y, \theta v) \in U$$

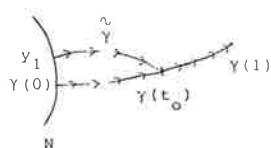
$$(6.8) \quad \mathcal{H}_1 \Big|_U \text{ is a diffeomorphism onto } \mathcal{H}_1(U).$$

$(\mathcal{H}_1(U))$ is then an open neighborhood of $\{Y(t) ; 0 < t < 1\}$. This follows from Proposition 6.3., the implicit function theorem, and the fact that the restriction of $\mathcal{H}_1$ to $\{(Y(0), \theta \dot{Y}(0)) ; 0 < \theta < 1\}$ is injective. We have the following known result :

Theorem 6.4. Under the assumptions of Proposition 6.3, we can find U as above, so that the function $\psi(x) = d(x, N)$ is C^∞ on $\mathcal{H}_1(U)$ and so that for every $x = \mathcal{H}_1(y, v) \in \mathcal{H}_1(U)$, the distance $\psi(x)$ is realized (up to reparametrisation) by a unique geodesic from N to x , namely $\gamma_x : [0,1] \ni t \rightarrow \mathcal{H}_1(y, tv)$.

Proof : We put $\tilde{\psi}(x) = \|v\|$ for $x = \mathcal{H}_1(y, v)$, $(y, v) \in U$. Then $\tilde{\psi}(x)$ is the length of the geodesic γ_x and $\tilde{\psi}$ is a C^∞ function on $\mathcal{H}_1(U)$. We shall prove that if U is small enough then $\tilde{\psi}(x)$ is the distance from x to N and that γ_x is the unique minimal geodesic from N to x . In fact, if no such U can be found then there is a sequence $x_j = \mathcal{H}_1(y_j, v_j)$ with $y_j \rightarrow Y(0)$, $v_j \rightarrow t_0 \dot{Y}(0)$, $0 < t_0 < 1$ such that N is joined to x_j by a minimal geodesic $\tilde{\gamma}_{x_j} : [0,1] \rightarrow M$ which is different from γ_{x_j} . We have $(\tilde{\gamma}_{x_j}(0), \dot{\tilde{\gamma}}_{x_j}(0)) \in TN^{\perp}$ and by Proposition 6.3., it is clear that there is a fixed neighborhood W of $(Y(0), t_0 \dot{Y}(0))$ in $TN^{\perp}$ where $(\tilde{\gamma}_{x_j}(0), \dot{\tilde{\gamma}}_{x_j}(0))$ cannot penetrate. Hence after passing to a subsequence, we have $(\tilde{\gamma}_{x_j}(0), \dot{\tilde{\gamma}}_{x_j}(0)) \rightarrow (y_1, v_1) \neq (Y(0), \dot{Y}(0))$, and if $\tilde{\gamma}$ is the corresponding limiting geodesic, then $\tilde{\gamma}(0) = y_1$, $\tilde{\gamma}(1) = Y(t_0)$. The length of $\tilde{\gamma}$ is at most equal to the length of $\gamma \Big|_{[0, t_0]}$ with is to $\|\dot{Y}(0)\|$ and by the minimality assumption we have equality. Then $\tilde{\gamma}(1) \neq t_0 \dot{Y}(t_0)$ while the two vectors have the same norm.

We are then in the case when N is joined to $\gamma(1)$ by two different curves of minimal length :



However, the curve given by $\tilde{\gamma}$ and $\gamma|_{[t_0, 1]}$ has a "corner" at $\gamma(t_0)$ and can therefore not be minimal. Hence by contradiction, we have proved Theorem 6.4.

#

We now return to the situation of section 5. Let V be non-negative and assume that $U_j = \{x_j\}$, $j = 1, \dots, N$ are points for which (5.14) hold. We do not assume (5.26)-(5.28). M is now as in the earlier sections and the discussion above will be applied to $M \setminus \{x_1, \dots, x_N\}$ equipped with the Agmon metric $V(x)dx^2$. Let $S_0 = \min_{j \neq k} d(x_j, x_k)$. For $1 \leq j \leq N$, let Ω_j consist of x_j and the union of the interiors of all minimal geodesics from x_j to some point in M , of length less than S_0 . Here, strictly speaking, we consider geodesics of the form $\gamma : [0, a] \rightarrow M \setminus \{x_1, \dots, x_N\}$, with $\gamma(t) \rightarrow x_j$ as $t \rightarrow 0$. In the earlier sections it was convenient to consider reparametrized geodesics defined on $]-\infty, \tilde{a}]$. A geodesic of the form γ above is nothing but the space projection of an integral curve of the Hamilton field $\frac{1}{V} \xi^2$. After reparametrization we may assume that this integral curve is in $\frac{1}{V} \xi^2 = 1$ and after another reparametrization we may consider γ as the projection of a (nul)-bicharacteristic of $q = \xi^2 - V(x)$.

Proposition 6.5. *The sets Ω_j are open and satisfy (5.11), (5.12) (with $0 = x_j$).*

Proof : For $\delta > 0$ small enough, we know that $\overline{B(x_j, \delta)} \subset \Omega_j$ and that there is a suitable neighborhood $\tilde{\Omega}_j \subset \Omega_j$ of this ball for which (5.11), (5.12) are satisfied. Then we only have to consider $\Omega_j \setminus \overline{B(x_j, \delta)}$. Let $N = \partial B(x_j, \delta)$. A minimal geodesic from x_j to a point $y \in B(x_j, S_0) \setminus B(x_j, \delta)$ decomposes into a minimal geodesic from x_j to N and one from N to y . Also, a minimal geodesic from N to such a point y can be extended to a minimal geodesic from

x_j to y . Clearly $d(y, x_j) = \delta + d(y, N)$. It is now clear that Theorem 6.4. implies that Ω_j is open, and that the distance to x_j is a smooth function $\psi_j(x)$ on Ω_j . After reparametrization we also know that the minimal geodesic from x_j to $x \in \Omega_j$ is the projection of the bicharacteristic $]-\infty, 0] \ni t \rightarrow (x(t), \xi(t))$ with $(x(t), \xi(t)) \rightarrow (x_j, 0)$ $t \rightarrow -\infty$, $(x(0), \xi(0)) = (x, \psi_j(x))$. This gives (5.12) and the proof is complete.

#

We choose M_j as in section 2 and an interval $I(h)$ as in section 5. Hence all the eigenvalues μ_α are asymptotically equal and we assume (5.29) in the case there are more than one μ_α associated to a given M_j . Let U_j, U_k satisfy $d(U_j, U_k) = S_0$. We choose Γ as in Remark 2.13. Then we have (5.31) if $j(\alpha) = j$, $j(\beta) = k$ and $\varphi_\alpha, \varphi_\beta$ admit asymptotic expansions near the intersection of Γ and all the minimal geodesics from U_j to U_k . After decreasing Γ if necessary we then still have (5.32), (5.33), so clearly the interaction coefficient $\hat{w}_{\alpha, \beta}$ will have an asymptotic expansion in many cases even without the assumptions (5.26), (5.28). In the case of the lowest eigenvalues, we get

Theorem 6.6. *Choose $I(h)$ as in Theorem 5.9 and assume that E_0 can appear in the spectrum of the various localized harmonic oscillators, only as the lowest eigenvalue (so there is no need for the assumption (5.29)). If $\mu_\alpha, \mu_\beta \in I(h)$ are eigenvalues of $P_{M_j(\alpha)}, P_{M_j(\beta)}$, with $d(U_j(\alpha), U_j(\beta)) = S_0$, then there is a constant $C > 0$ such that*

$$\frac{1}{C} h^{1/2} \leq (-\hat{w}_{\alpha, \beta}) e^{S_0/h} \leq C h^{1-n/2}.$$

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NON-NEGATIVE SOLUTIONS OF THE POROUS MEDIUM EQUATION

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1. Introduction

The purpose of this paper is to study properties of solutions of the equation

$$(1.1) \quad \frac{\partial u}{\partial t} = \Delta u^m, \quad m > 1, \quad u \geq 0.$$

This equation arises as a model for gas flow in a porous medium; the solution $u(x,t)$ represents the density of the gas. The equation (1.1) is a non-linear, degenerate parabolic equation, and has been studied extensively. For a recent survey, see [9].

In this work we study nonnegative continuous weak solutions of (1.1) in a strip $S_T = \{(x,t): x \in \mathbb{R}^n, 0 < t \leq T\}$. That is, we consider continuous functions $u \geq 0$ in S_T , for which